Taxation when markets are not competitive: Evidence from a loan tax

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Abstract

We examine how lender conduct shapes the incidence and welfare effects of financial taxes and subsidies. Exploiting Ecuador's unanticipated 2014 loan tax and administrative lending data, we estimate and test a structural model nesting Bertrand-Nash competition, joint maximization, and partial cartels, identified using tax pass-throughs. We reject the Bertrand-Nash benchmark: banks internalize rivals' profits when setting prices. Imposing Bertrand-Nash pricing overstates tax incidence on borrowers, understates fiscal capacity, and misstates the excess burden in a direction that depends on whether forgone supra-competitive profits count as a social loss. Conduct is therefore a first-order input to evaluating financial taxes and subsidies.

JEL Classification Codes: Banks (G21); Government Regulation of Banks (G28); Taxation and Subsidies (H2); Market Structure, Firm Strategy, and Market Performance (L1)

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Governments frequently levy taxes on bank loans and subsidize credit. These instruments are attractive because banks allocate savings across the economy and loans constitute a well-monitored tax base. As a result, the taxation and subsidization of credit is a persistent focus of public debate. Yet this debate often abstracts from lender conduct—the degree to which banks' pricing accounts for rivals' responses—even though theory shows that conduct shapes economic incidence and welfare outcomes (Weyl and Fabinger (2013)). In practice, empirical studies of credit markets often assume a high degree of competition. While useful as a benchmark, lending markets are often far from this ideal. Banks typically have pricing power and interact strategically, departing from the competitive environments of the classical tax-incidence literature (Slemrod (1990); Auerbach (2002)).[1] As a result, assumptions about how lenders compete may have first-order implications for evaluating who bears the burden of financial taxes and subsidies. This paper addresses that issue by testing the extent to which lenders internalize the effects of their pricing decisions on rivals' profits, and by quantifying how competitive conduct shapes the incidence and welfare effects of financial taxes and subsidies.[2]

Our study makes two empirical contributions. First, we are the first to test conduct using modern industrial organization tools in a market with selection.[3] A key challenge in testing conduct with selection is that standard instruments are demand-side variables that shift markups through demand elasticities, but in selection markets these same variables also move banks' costs through changes in the borrower pool, violating the exclusion restriction. We propose a financial tax as an alternative source of markup variation. It shifts markups through lender costs without directly impacting credit demand, satisfying the exclusion restriction. Using this framework, we reject the standard benchmark of differentiated-product price competition (Bertrand-Nash). Second, we quantify the consequences of imposing incorrect conduct assumptions. We show that misspecifying how lenders compete leads to substantial errors in measured tax incidence and welfare effects of financial taxes and subsidies. For policy evaluation and credit-market modeling alike, lender conduct assumptions are first order.

Our paper focuses on the unanticipated reintroduction in 2014 of a loan tax in Ecuador to fund a public cancer hospital (Sociedad de Lucha Contra el Cáncer, or SOLCA). The SOLCA

[1] For empirical evidence on competition in banking and its consequences, see Cornaggia et al. (2015); Hatfield and Wallen (2023); Brugués et al. (2023); and Jiang et al. (2023).

[2] Throughout the paper, we use terms such as "profit internalization" and "joint maximization" to describe pricing behavior that departs from Bertrand-Nash competition. These terms refer to economic behavior rather than to an identified legal finding of collusion.

[3] By selection, we mean that borrower characteristics that affect credit demand also affect banks' costs. This arises through two channels. First, as the composition of the borrower pool changes over time, so do average costs of lending. Second, borrower characteristics such as default risk affect both the cost of serving a loan and the borrower's willingness to pay (as in Benetton et al. (2025); Einav et al. (2021)). We do not model asymmetric information; all selection in our framework operates through observable borrower characteristics.

tax is a one-time charge levied at loan approval, equal to about 0.5% of the face value of credit.[4] The tax was unanticipated and swiftly implemented (introduced in September, in effect by October 2014), leaving little scope for anticipatory adjustment by lenders or borrowers. The reform, therefore, provides a clean setting for studying lending markets' responses to financial transaction taxes, a policy instrument used in more than 40 countries.[5] Moreover, the tax constitutes a plausibly exogenous shock to the marginal cost of new lending, orthogonal to contemporaneous demand shifts. We study this shock using administrative data on the universe of commercial bank lending in Ecuador from 2010 to 2017.[6] It includes detailed contract terms, loan performance, and borrower balance sheets, which allows us to track how loan prices adjust around the tax introduction while holding borrower risk and relationship characteristics fixed.

To study the effects of taxation under alternative modes of competition, we propose a supply-side generalization of the workhorse credit demand-and-supply model (e.g., Crawford et al., 2018; Benetton, 2021). The model features discrete-continuous demand and incorporates the internalization approach of Miller and Weinberg (2017) to nest Bertrand-Nash, partial cartels, and full joint maximization. On the demand side, borrowers choose which bank to borrow from, or whether not to borrow at all. The discrete-choice demand system is flexible, with random coefficients on constants and prices, as well as price-demographic interactions. Conditional on borrowing, firms then choose how much to borrow, again allowing price sensitivities to vary with demographics.

On the supply side, banks choose prices to maximize profits, subject to an internalization parameter that measures the extent to which they weight competitors' profits in their own pricing decisions. When pricing, banks face two types of observable selection: (1) heterogeneous default risk and (2) heterogeneous marginal costs. The conduct-testing framework developed in Backus et al. (2024), Duarte et al. (2024), and Dearing et al. (2024) can be adapted to this setting with selection. However, traditional instruments based on borrower demographics and product characteristics (Backus et al. (2024); Duarte et al. (2024)) are unlikely to satisfy the exclusion restriction (Einav et al. (2021)), because marginal costs, default risk, and demand comove. Instead, we rely on taxation as observable cost shocks that shift markups, making them valid instruments for testing (Berry and Haile (2014)). The validity of tax-based instruments requires that the tax change be unexpected (exogeneity) and generate positive pass-through to final interest rates (relevance).

[4]The tax rate varied by maturity for short-term credit during our sample period (Section 1.1).

[5]Centre for Economic Policy Research estimates (https://cepr.net/report/financial-transactions-taxes-around-the-world/, accessed November 1, 2024).

[6]Ecuador is a bank-dependent economy where 60% of firms report having bank credit (World Bank Enterprise Survey, 2017), while access to public equity and bond markets is negligible.

We validate these testing requirements using an event-study design controlling for firm and bank fixed effects. For private banks, we document a sharp and persistent response of contracted loan rates to the tax. Borrowers' all-in cost of credit (interest rate plus tax) rises by roughly one-third to one-half of the tax amount on average, while banks absorb the remainder through lower pre-tax rates. This incomplete pass-through also holds within active firm-bank relationships, ruling out compositional changes in borrower risk as the driver. The price response is absent for government-owned banks that are exempt from the tax, supporting the interpretation of the reform as a plausibly exogenous cost shock. The absence of pre-trends validates the claim that the tax was unexpected. The results hold when relying on a difference-in-differences approach using private banks as treatment and government banks as control. Moreover, we show that the result is robust to controls for potential confounders, including oil shocks and deposit rates. Lastly, the pass-through remains non-zero even after controlling for bank-quarter fixed effects, indicating that tax variation induced by differences in contracted maturities implies differential post-tax interest rates.

Beyond validating the testing requirements, the literature has argued that persistent incomplete pass-through is prima facie evidence of imperfect competition (e.g., Drechsler et al. (2017); Benetton and Fantino (2021); Eisenschmidt et al. (2024)). We show that pass-through varies systematically with proxies for competition, such as the number of potential lenders, and with proxies for potential coordination, such as multi-market contact indices. While suggestive, these reduced-form patterns are not sufficient to discipline the supply-side model. We therefore use an alleged cartel structure to guide our empirical application. In particular, we study the private banking association (ASOBANCA or BA) as a potential focal point, given concerns raised by the antitrust authority regarding anti-competitive behavior.[7] In our setting, pass-through becomes less complete as the BA becomes more relevant in a given market.

Reduced-form evidence alone cannot fully characterize welfare, since the same pass-through rate is consistent with multiple market structures (Weyl and Fabinger (2013)). For that reason, we estimate demand using the flexible discrete-continuous framework proposed above. The model fits the data well and yields sensible relationships between variables. For example, firms are more likely to choose banks with longer relationships, while older firms are less likely to borrow but, conditional on borrowing, take larger loans. Importantly, the estimated demand accommodates a wide range of elasticities that vary with borrower characteristics and can generate both incomplete and more-than-complete pass-through.

With estimated demand in hand, we implement the conduct-testing framework (Backus et

[7]Details on BA's institutional role, antitrust record, and reduced-form evidence linking membership to lower pass-through are provided in Section 3.3 and Internet Appendix E.

al. (2024); Duarte et al. (2024); Dearing et al. (2024)) and reject Bertrand-Nash competition (under which banks behave as monopolists over their residual demand curves) at conventional significance levels. Instead, the data are better explained by pricing behavior in which banks internalize rivals' profits within subsets of lenders. We consider three alternative internalization structures: (1) an alleged BA cartel, (2) an alleged top-4 bank cartel, and (3) a full bank cartel (joint maximization). Among these candidate coalition structures, profit internalization based on BA membership or the top 4 banks (all members of the BA) are most consistently supported.[8] We also conduct out-of-model validation exercises using recovered marginal costs. In particular, marginal costs under the alleged cartels align more closely with deposit rates (a model-free proxy for the marginal cost of lending) than Bertrand-Nash. Moreover, recovered marginal costs are stable over time, suggesting that our testing results are not driven by errors in cost recovery from demand estimates and further validating the best-fit market structure (Bresnahan, 1987).

Although the flexible demand system reduces the risk of false rejections due to demand misspecification, conduct tests could still be biased if banks face economies or diseconomies of scale. Because a single tax introduction cannot separately identify conduct and scale economies (Duarte et al. (2025)), we apply separate tests of marginal-cost curvature under each conduct model considered. For each model, we fail to reject constant marginal costs, reducing concerns that our testing results are driven by supply-side misspecification arising from scale economies.

Having empirically rejected the Bertrand-Nash benchmark commonly used in the literature, we use the model to simulate the welfare consequences of the SOLCA tax under the conduct models that our tests cannot reject. Market structure substantially alters how distortionary loan taxes are. Under Bertrand-Nash, borrowers and banks share the tax burden roughly equally, and the policymaker anticipates an excess burden of \$0.82 per dollar of revenue raised. When banks fully internalize rivals' profits within the alleged cartels, the burden shifts toward banks and the ratio of borrower to bank losses falls from 0.99 to 0.87 under the Top-4 cartel and 0.67 under the BA cartel. Banks lose more because the tax erodes the supra-competitive margins that internalization creates. Whether that makes the tax more or less distortionary depends on how the policymaker values those margins. Counted as a social loss, the excess burden rises from \$0.82 per dollar of revenue under Bertrand-Nash to \$0.90 and \$1.10 under the Top-4 and BA cartels. Given no social weight, because taxes in less competitive markets partially extract existing rents rather than reducing lending, it instead falls to \$0.79 and \$0.72, respectively, up to 12% below Bertrand-Nash. Under weak internalization, these outcomes remain close

[8] Our objective is to test whether pricing behavior departs from standard competitive benchmarks, not to establish explicit collusion, which would require different data and analyses.

to Bertrand-Nash. These effects vary systematically, though modestly, by firm size: borrower incidence increases with firm size under Bertrand-Nash but decreases with firm size under full BA internalization, highlighting how policymakers may reach different conclusions depending on their supply-side assumptions.

Lender conduct also shapes fiscal capacity and subsidy effectiveness. Less competitive pricing shifts the Laffer curve outward, consistent with Miravete et al. (2018), raising revenue at a given tax rate and increasing the revenue-maximizing rate. For instance, at revenue maximizing rates, tax revenue would be 5 to 6% higher under the cartels than Bertrand-Nash. Finally, the same mechanism that operates for taxes works in reverse for subsidies: banks capture a larger share of the benefit, making subsidies less effective at expanding borrowing. Together, these results show that competition in lending markets governs both who bears the burden of financial taxes and how effective tax and subsidy policies are.

Our study focuses on the 2014 SOLCA tax in Ecuador, but its implications extend broadly. Concentrated banking sectors and financial transaction taxes and subsidies are common worldwide.[9] More broadly, our approach extends to any setting where a quasi-exogenous cost shock can identify how conduct mediates the incidence and welfare effects of regulation, such as studying monetary policy pass-through or other policy interventions in imperfectly competitive lending markets.

Literature Review. We contribute to the literature on cost pass-through and tax incidence. That market structure affects the incidence and deadweight loss of taxation has been recognized since Harberger (1962) and was formalized for imperfectly competitive markets by Weyl and Fabinger (2013) (see also Nakamura and Zerom (2010) and Fabra and Reguant (2014)). Recent work empirically studies how pass-through varies with regulatory or demand-based drivers of market power (Cabral et al. (2018); Ganapati et al. (2020); Conlon and Rao (2026); O'Connell and Smith (2024)). Closest to our paper, Cabral et al. (2018) study pass-through in health insurance, also a selection market, and decompose pass-through into selection, risk, and demand-based market power. However, they do not focus on supply-side drivers of market power. To our knowledge, ours is the first paper to study supply-side drivers of pass-through while accounting for selection and risk, and the first to quantify how market structure mediates tax incidence and welfare in a selection market.

Second, we contribute to the banking literature by testing alternative models of lender conduct rather than assuming one *a priori*. Much of the lending literature assumes Bertrand-Nash

[9]We document that the Ecuadorian commercial lending market is broadly representative in Section 1 and Internet Appendix A. We benchmark our estimates against the literature throughout the paper.

pricing and explains markups through search frictions or borrower heterogeneity, including in commercial lending (Crawford et al. (2018); Cox et al. (2026)), deposits (Egan et al. (2017)), mortgages (Allen et al. (2019); Robles-Garcia (2021); Benetton (2021)), auto loans (Yannelis and Zhang (2023)), and consumer credit (Cuesta and Sepúlveda (2024)). One exception is Hatfield and Wallen (2023), who consider bank coordination through multi-market contact but do not estimate a flexible conduct model.[10] Relative to this literature, we generalize bank conduct to allow markups to reflect both demand-side and supply-side sources of market power.

Third, we contribute a methodological innovation for conduct testing in selection markets, with applications beyond banking. Following Bresnahan (1982), the industrial organization literature uses exclusion restrictions from plausibly exogenous markup shifters to test alternative conduct models. Recent examples (Nevo (2001); Miller and Weinberg (2017); Backus et al. (2024); Calder-Wang and Kim (2024)) rely on demand-side instruments that shift markups without affecting marginal costs (Berry and Haile (2014); Backus et al. (2024); Duarte et al. (2024); Dearing et al. (2024)). In markets with selection, however, demand-side instruments may also correlate with marginal costs, violating the exclusion restriction.[11] We instead use tax-induced pass-through as identifying variation to test competing conduct models. Our approach parallels Delipalla and O'Donnell (2001), who use cigarette tax changes to test conduct, and Rojas (2008), who exploit beer tax variation. Contemporaneously, Dearing et al. (2024) show theoretically and empirically that pass-through is the economic determinant of instrument relevance for conduct testing. Our methodology extends the classic approaches of Sumner (1981) and Sullivan (1985), and is in the spirit of Atkin and Donaldson (2015), who use observable pass-through to determine surplus division in international trade, and Bergquist and Dinerstein (2020), who use experimentally estimated pass-throughs in agricultural markets to test for coordination. Our approach can be applied broadly in selection markets wherever quasi-experimental cost variation is available, including studies of monetary policy pass-through, regulatory interventions, and supply-chain pricing.

Finally, our paper relates to the finance literature on monetary policy pass-through to loan and deposit rates (Di Maggio et al. (2017); Drechsler et al. (2017); Benetton and Fantino (2021); Wang et al. (2022); Eisenschmidt et al. (2024); Li et al. (2026)). Benetton and Fantino (2021) and Wang et al. (2022) find that monetary policy pass-through to loan rates is more complete in competitive markets. Drechsler et al. (2017) show that federal funds rate increases are not fully passed through to deposit rates, especially in concentrated markets, while Hatfield and Wallen

[10] Outside lending, Ciliberto and Williams (2014) study coordination via multi-market contact in airlines but also do not estimate flexible conduct parameters.

[11] For example, firm growth rates, assets, and CEO age shift demand but are also linked to risk and thus affect the marginal cost of lending.

(2023) show that multi-market contact reduces deposit-rate pass-through. We emphasize that market power can also arise from supply-side softening of price competition, not only from demand-side sources.[12] Because our findings show that conduct assumptions are first order for both the magnitude and distributional effects of policy interventions, our methodology could discipline those assumptions in monetary policy settings. In Ecuador, for example, our findings suggest that expansionary monetary policy may produce more muted credit responses than standard models predict, since banks with pricing power internalize part of the subsidy.

The rest of the paper is organized as follows. Section 1 describes the SOLCA tax, the Ecuadorian credit market, and the data. Section 2 presents the lending model, with Subsection 2.3 describing how we identify the internalization conduct parameters using pass-through from the SOLCA tax. Section 3 estimates tax pass-through and validates the identification assumptions. Section 4 estimates the demand model and presents model fit. Section 5 presents the testing strategy and results for the supply model. Section 6 presents welfare analysis of the SOLCA tax and how it depends on lender competitive conduct. Section 7 concludes.

1 Description of the SOLCA Tax and Dataset

1.1 Loan Tax

Like many developing countries, particularly in Latin America, Ecuador employs financial transaction taxes as a revenue source (Kirilenko and Summers (2003)). From 1964, Ecuador utilized a bank levy to fund free cancer treatment centers run by the Sociedad de Lucha contra el Cáncer (SOLCA). These taxes applied to all financial operations at rates between 0.25 to 1 percent of the value of the transaction or loan. In 2008, the Ecuadorian government eliminated all taxes on financial transactions, opting instead to fund SOLCA through the regular budget.

The re-introduction of the SOLCA tax was unanticipated and implemented swiftly. In September 2014, the Ecuadorian National Assembly passed the legislative framework governing the country's financial and monetary system, the "Código Orgánico Monetario y Financiero" to consolidate existing regulations in one framework and to pass new regulation of mobile money payments and strengthen anti-money laundering measures.[13] A last-minute

[12] In a companion paper, Brugués et al. (2023), we quantify the direct welfare and aggregate impacts of imperfect competition.

[13] In civil law countries like Ecuador, such codes are comprehensive legal statutes that centralize and standardize regulation for all institutions and activities related to finance and monetary policy, akin to a "financial constitution." The code delineates regulatory authority, outlines obligations and operational rules for banks, insurance companies, and other financial entities, and specifies consumer protection standards.

amendment reintroduced a SOLCA tax to address funding gaps for cancer treatment. This reintroduction was unexpected by both borrowers and financial institutions in Ecuador.[14] The government implemented the new SOLCA tax by the end of October 2014, a mere month after the law's passage.

The tax, collected by banks at loan grant and remitted to the tax authority, is levied on borrowers for each new loan. It applies to commercial, credit card, auto, and mortgage loans. Throughout our sample period, 2010 through 2017, only loans from private banks were subject to the SOLCA tax; the law exempted loans from government banks. In practice, as government banks have a small market share in the conventional commercial credit market, the tax applied to most commercial loans.[15] The tax amount varies with loan maturity: loans with a one-year or longer term incur the full 0.5% tax, while shorter-term loans are taxed proportionally.[16] Next, we describe the data that allow us to pin down the impact of the tax on commercial loan terms.

1.2 Datasets

We construct a comprehensive dataset by combining administrative databases collected by Ecuador's bank regulator, the Superintendencia de Bancos (Superbancos), and its business bureau, the Superintendencia de Compañías (Supercias). The data are quarterly and span the period between January 2010 and December 2017.

The primary data are the universe of new and outstanding commercial bank loans from banks operating in Ecuador between 2010 and 2017. This encompasses loans from 23 private and 4 government banks. The dataset is not a credit registry since it does not allow banks to view other banks' loan information. But it provides similar types of information. Variables include loan amount, type, interest rate, term-to-maturity, and internal bank risk assessments at the time of loan issuance. Additionally, it includes quarterly snapshots on outstanding loans, including repayment performance data and loan drawdown. Loan performance data allow us to observe default; we define a loan as in default when the borrower is behind on payments for 90 days or more.

For our analyses, we focus only on regular commercial loans issued to corporations reg-

[14] See, for example, contemporary coverage in the two major Ecuadorian newspapers: "Código revive impuesto de 0,5% para créditos para beneficiar a SOLCA," by the editorial staff, published the 29th of July 2014, in *El Universo*; and "El Código Monetario pasó con reformas de última hora," by Mónica Orozco, published the 25th of July 2014 in *El Comercio*.

[15] See Appendix A for a description of public and private commercial lending, the extent to which these markets were segmented, and how the tax affected this segmentation.

[16] The tax rate for loans with maturities less than one year is calculated as $0.5\% \times X/12$, where X is the loan's maturity in months.

ulated by Ecuador's business bureau ("SA" firms, for "Sociedad Anónima").[17] This excludes microloans and loans to sole proprietorships. These filters match the coverage of our firm-level data and allow us to specialize our model of commercial credit. For example, market entry and competition within the microlending sector differ considerably from commercial lending by private banks. We use a unique firm identifier to merge the loan dataset with annual, firm-level data from the Supercias, Ecuador's business bureau. This dataset provides balance sheets, income statements, and wage information. The combined firm observables and realized default allow us to construct a default prediction model at loan origination, which we describe in Internet Appendix C.3.

1.3 Descriptive Statistics

We now describe the Ecuadorian commercial loan market and our main variables. Table 1 displays bank-province-year level credit statistics. The average (median) bank issues \$57M (\$1.4M) in corporate loans in each province-year. The average bank lends to 83 corporations in a given province-year, but there are also bank-provinces with few commercial clients, as the median bank has 12 firm clients a province-year. In total, banks offer 487 (24) loans in an average (median) province-year. These patterns in the competitive structure of Ecuador's commercial loan sector are representative of corporate lending elsewhere, especially in bank-based financial systems, where it is common to observe a few dominant national banks lending alongside smaller banks specializing regionally or in particular loan segments.[18]

[Place Table 1 here.]

To test if the Ecuadorian commercial market relates to bank competition similarly to what has been documented in the banking literature, Table 2 presents descriptive statistics on market access and lending by market concentration, as measured by the Herfindahl–Hirschman Index (HHI) based on commercial lending share over 2010 to 2017. Sensibly, highly-concentrated markets feature fewer branches and fewer competing banks. Branches in highly-concentrated markets are smaller, cater to fewer clients, offer fewer loans in total and per client, and offer slightly shorter maturities and charge higher interest rates.

[17] In Ecuador, business structures include stock corporations (SA) and limited liability companies (SL). The key difference is that shares in stock corporations can be freely traded, while quotas in limited liability companies require unanimous consent to transfer.

[18] Internet Appendix Figure A2 confirms that Ecuador's three-bank concentration ratio is near the LAC median. The coexistence of a few large banks exercising market power alongside a competitive fringe of smaller lenders is a feature commonly explored in empirical banking (e.g., Drechsler et al. (2017)).

[Place Table 2 here.]

Next, Table 3 describes the merged commercial loan dataset spanning the years 2010 to 2017. Panel A summarizes the data at the firm-year level. We have 452,486 firm-year observations, corresponding to 30,996 unique corporations. Of these, 92,659 firm-year observations relate to active firm-year borrowers, whereas 359,827 observations pertain to non-borrowing firm-years. The average borrowing firm is roughly twelve years post-incorporation and possesses, on average, $2 million in assets. The median firm holds $400,000 in total assets, indicating that, as is common elsewhere, the firm size distribution is skewed. Total sales demonstrate a similar skew, with average (median) sales of $2.4 million ($600,000). The average (median) borrowing firm is highly leveraged, displaying an average (median) debt-to-assets ratio of 0.67 (0.71). On average (at median), firms maintain 1.4 (1) banking relationships within a given borrowing year. Most clients repeatedly borrow, as indicated by the average client borrowing 9 (2) times in a year.

[Place Table 3 here.]

In contrast, Panel B reports that non-borrowing firms are generally younger, with a mean age of around ten years since incorporation, and are smaller, with mean (median) assets of $460,000 ($50,000) and mean (median) sales of $430,000 ($30,000). These non-borrowing firms are also less leveraged, with a total debt-to-assets ratio of 0.54 (0.58).[19] Panel C describes data at the bank-firm-year level. The average (median) duration of borrower-lender relationships is 2.29 (2) years. Panel D summarizes loan-level data for the universe of commercial (non-micro) loans granted to corporations between 2010 and 2017. Each loan is for an average (median) of $70,000 ($10,000) and the average (median) term-to-maturity is six (three) months. The average (median) annualized nominal interest rate is 9.83% (9.02%). Finally, Panel E reports average bank-year level deposit rates, a proxy for the marginal cost of funds, of 4.7%.

The banks in our sample only write down the value of about two percent of the loans that they issue. Actual default is a rare occurrence in our sample, happening less than one percent of the time. In the total sample, which includes sole proprietorships and micro-loans, the default rate is three percent. The combination of low default rates, high lending rates, and a moderate level of marginal cost based on the deposit rate proxy is indicative of lender market power.

[19] Non-borrowing firms may have leverage from sources other than traditional banks. For instance, micro-loans from micro-lending institutions and trade-credit.

To speak about the generality of our findings, we provide additional evidence that the Ecuadorian commercial loan market is representative in Internet Appendix A, where we benchmark Ecuador against other LAC and emerging-market economies on several banking-sector indicators and report correlations between average equilibrium interest rates and market characteristics at the aggregated bank-province-year level. As with the competitive structure of the commercial loan market, the general patterns we observe between market access and loan pricing align with those documented in the existing banking literature (Hannan (1991); Beck et al. (2013); Van Leuvensteijn et al. (2013)). The loan pricing regularities we document, including interest rate dispersion among similar borrowers and spreads that decline with loan size and relationship length, have also been found in commercial lending markets in the U.S. (Amiti et al. (2026)), Spain (Martín-Oliver et al. (2008)), France (Mazet-Sonilhac (2025)), and Chile (Bordeu et al. (2025)). These parallels concern the features of the lending environment rather than our conduct and pass-through results, which are estimated for Ecuador alone.

2 Commercial Lending Model with Competitor Internalization

In this section, we introduce the theoretical framework: a quantitative model of commercial lending with competitor internalization. The model enables us to directly characterize bank competition and its impacts. Our model is most applicable to small-to-medium-sized, single-establishment firms and to private, traditional, deposit-funded banks. We assume that borrowers and lenders are risk neutral, borrowers have the freedom to choose from any bank in their local market (or to not borrow at all), and the returns on borrowers' investments can be parameterized.

2.1 Demand side

First, each firm i is classified into a group $g(i)$ based on observable product and firm characteristics (e.g., SME classification). Each firm i in period t decides whether to borrow from a bank $k \in K_{g(i)mt}$, the set of active banks in local market m offering credit at interest rates $r_{g(i)kmt}$ to borrowing group $g(i)$.[20] We assume that the indirect profit that firm i derives from choosing bank k is:

$$\Pi^s_{ikmt} = \delta^s_{g(i)kmt} + \alpha^s_{ikmt} r_{g(i)kmt} + \beta^s_1 X_{it} + \beta^s_2 X_{ikmt} + \psi^s_{it} + \varepsilon^s_{ikmt}. \tag{1}$$

[20] Most borrowers have only one lender at a given point in time (see Table 3).

Here, $\delta^s_{g(i)kmt}$ captures time-varying preferences for banks that are common to all firms in the group. Borrower-specific price sensitivity is parameterized as $\alpha^s_{ikmt} = \alpha^s + \alpha^{s,x}_1 X_{it} + \alpha^{s,x}_2 X_{ikmt} + \tilde{\alpha}^s_{it}$, where α^s is the mean sensitivity, $\alpha^{s,x}_1$ and $\alpha^{s,x}_2$ capture heterogeneity driven by observables, and $\tilde{\alpha}^s_{it}$ is an unobserved random coefficient. The vector X_{it} denotes observable firm characteristics, such as assets and revenue, which affect the likelihood of borrowing, while X_{ikmt} represents time-varying attributes of the bank-firm relationship, such as relationship length, which affect the likelihood of the firm choosing a specific bank. Finally, ψ^s_{it} captures unobserved firm-time demand shocks, similar to a pseudo-nest between borrowing or not borrowing, and ε^s_{ikmt} is an idiosyncratic error term.

If the firm chooses not to borrow, it receives its outside option with indirect profit $\Pi^s_{i0mt} = \varepsilon^s_{i0mt}$, with the deterministic portion of indirect profit normalized to zero. Firm i selects bank k that maximizes its expected indirect profit, yielding choice probability $s_{ikmt} = Prob(\Pi^s_{ikmt} \geq \Pi^s_{ik'mt}, \forall k' \in K_{g(i)mt} \cup \{0\})$.

Conditional on choosing to borrow from bank k, firm i also chooses how much to borrow. Total expected loan demand from firm i for bank k is therefore $Q_{ikmt}(r) = s_{ikmt}(r)L_{ikmt}(r)$, the product of the probability of choosing bank k and the expected loan size conditional on that choice and given posted interest rates $r_{g(i)mt} = \{r_{g(i)kmt}\}_{k \in K_{g(i)mt}}$. We model the loan size decision using a separate indirect profit function

$$\Pi^l_{ikmt} = \delta^l_{g(i)kmt} + \alpha^l_{ikmt} r_{g(i)kmt} + \beta^l_1 X_{it} + \beta^l_2 X_{ikmt} + \varepsilon^l_{ikmt}, \tag{2}$$

where the terms parallel those in the discrete choice equation. Here, $\delta^l_{g(i)kmt}$ captures cross-sectional differences in loan use across banks, α^l_{ikmt} captures heterogeneous price-sensitivities such that $\alpha^l_{ikmt} = \alpha^l + \alpha^{l,x}_1 X_{it} + \alpha^{l,x}_2 X_{ikmt}$, which allows for loan use sensitivity to depend on relationship and firm characteristics. Besides price sensitivities, the loan demand indirect utility also accounts for heterogeneous needs for loan amounts based on observable characteristics. Lastly, ε^l_{ikmt} captures idiosyncratic shocks.

2.2 Supply side

On the supply side, we model competition between banks using a flexible internalization framework that captures supply-side sources of market power. Intuitively, banks set interest rates to maximize an objective function that includes not only their own profits but also a weighted sum of their competitors' profits (Miller and Weinberg (2017)). This internalization captures a variety of competitive scenarios, including pure Bertrand-Nash competition (where each bank maximizes profits given its residual demand), partial cartels (where banks internalize the profits

of specific peers), and joint maximization (where banks jointly maximize industry profits).

Bank k sets the interest rate $r_{g(i)kmt}$ for all firms i in group g within market m at time t.[21] The bank's maximization problem is

$$\begin{aligned}\max_{r_{g(i)kmt}} B_{gkmt} = &\sum_{i\in g}\Big[(1-d_{ikmt})r_{g(i)kmt} - mc_{ikmt}\Big]\,Q_{ikmt}(\tilde{r}_{imt}) \\ &+ \sum_{i\in g}\sum_{j\neq k}\upsilon_{kj}\Big[(1-d_{ijmt})r_{g(i)jmt} - mc_{ijmt}\Big]\,Q_{ijmt}(\tilde{r}_{imt}),\end{aligned} \tag{3}$$

where υ_{kj} represents the internalization parameter, or profit weights, measuring the weight bank k places on bank j's profits. Given the institutional setting of a small economy with a single tax event, we assume the internalization parameters υ_{kj} are constant across borrower groups, markets, and time. The term d_{ikmt} denotes the firm's expected default probability at loan origination. $\tilde{r}_{imt} = \{\tilde{r}_{ikmt}\}_{k\in K_{g(i)mt}}$ is the tax-inclusive price vector, where $\tilde{r}_{ikmt} = r_{g(i)kmt} + \tau_{ikmt}$ and τ_{ikmt} is the tax shock, which takes values between 0 and 0.5, depending on loan maturity.

The first-order condition (FOC) for bank k with respect to the interest rate $r_{g(i)kmt}$ is:

$$\begin{aligned}&\sum_{i\in g}\left((1-d_{ikmt})Q_{ikmt}(\cdot) + \Big[(1-d_{ikmt})r_{g(i)kmt} - mc_{ikmt}\Big]\frac{\partial Q_{ikmt}(\cdot)}{\partial r_{g(i)kmt}}\right) \\ &\quad + \sum_{i\in g}\sum_{j\neq k}\upsilon_{kj}\left[((1-d_{ijmt})r_{g(i)jmt} - mc_{ijmt})\frac{\partial Q_{ijmt}(\cdot)}{\partial r_{g(i)kmt}}\right] = 0.\end{aligned} \tag{4}$$

The first line of Equation 4 is the standard pricing decision under Bertrand-Nash competition, capturing the direct effect of rate changes on margins and the indirect effect through the own-price elasticity of demand. The second line explicitly incorporates how bank k's pricing affects competitors' profits, weighted by the internalization parameters υ_{kj}. This formulation flexibly encompasses various market structures. For instance, when $\upsilon_{kj} = 0$, banks compete in a standard Bertrand-Nash fashion; when $\upsilon_{kj} = 1$, they act as a full cartel maximizing joint industry profits; and alternative arbitrary values of υ_{kj} capture partial cartel behavior and nuanced market conduct.

To facilitate the interpretation of Equation 4, we rewrite the first-order conditions using matrix notation and derive the pricing equation. We define the default-adjusted quantity vector $\mathbf{Q}^d_{imt} \equiv (1 - \mathbf{d}_{imt}) \circ \mathbf{Q}_{imt}$, where $\circ$ denotes the element-by-element product, and the default-adjusted matrix of demand derivatives with respect to prices, $\mathbf{\Delta}^d_{imt} \equiv \frac{\partial Q_{imt}}{\partial r'_{g(i)mt}}\,\mathrm{diag}(1 - d_{imt})$, and

[21] Internet Appendix Table C1 provides empirical support for this group-level pricing assumption. Pricing-cell fixed effects explain 85% of loan-level interest rate variation ($R^2 = 0.853$), and adding individual borrower characteristics raises the R^2 by only 0.003, offering support for a take-it-or-leave-it setting.

its unadjusted counterpart $\mathbf{\Delta}_{imt} \equiv \frac{\partial Q_{imt}}{\partial r'_{g(i)mt}}$. Finally, $\mathbf{\Upsilon}_{gmt}$ is the $K_{gmt} \times K_{gmt}$ internalization matrix, with ones on the diagonal and υ_{kj} on the off-diagonals. This yields first-order conditions

$$\sum_{i \in g} \Big([\mathbf{\Upsilon}_{gmt} \circ \mathbf{\Delta}^d_{imt}] \mathbf{r}_{g(i)mt} - [\mathbf{\Upsilon}_{gmt} \circ \mathbf{\Delta}_{imt}] \mathbf{mc}_{imt} \Big) = - \sum_{i \in g} \mathbf{Q}^d_{imt}. \tag{5}$$

Solving for prices yields a structural markup equation

$$\mathbf{r}_{gmt} = \widetilde{\mathbf{mc}}_{gmt}(\mathbf{\Upsilon}_{gmt}) + \boldsymbol{\mu}_{gmt}(\mathbf{\Upsilon}_{gmt}) \tag{6}$$

where the markup term

$$\boldsymbol{\mu}_{gmt}(\mathbf{\Upsilon}_{gmt}) = -\Big[\sum_{i \in g} \mathbf{\Upsilon}_{gmt} \circ \mathbf{\Delta}^d_{imt} \Big]^{-1} \sum_{i \in g} \mathbf{Q}^d_{imt} \tag{7}$$

depends on a vector of "virtual" elasticities, accounting for effective default rates and the degree of cross-profit internalization. The term

$$\widetilde{\mathbf{mc}}_{gmt}(\mathbf{\Upsilon}_{gmt}) = \Big[\sum_{i \in g} \mathbf{\Upsilon}_{gmt} \circ \mathbf{\Delta}^d_{imt} \Big]^{-1} \sum_{i \in g} [\mathbf{\Upsilon}_{gmt} \circ \mathbf{\Delta}_{imt}] \mathbf{mc}_{imt}. \tag{8}$$

represents a vector of "virtual" marginal costs. The virtual marginal cost relevant to bank k is a weighted average of two components: (1) mc_{ikmt} representing the bank's own cost of lending to borrower i, weighted by the own-price sensitivity $\frac{\partial Q_{ikmt}}{\partial r_{g(i)kmt}}$; and (2) competitors' costs (mc_{ijmt}), weighted by the internalization-adjusted cross-price sensitivity $\upsilon_{kj}(1 - d_{ijmt}) \frac{\partial Q_{ijmt}}{\partial r_{g(i)kmt}}$.

Intuitively, the weights reflect the constraints on bank pricing power. A higher sensitivity of borrower i to price changes constrains bank k's pricing more, thereby increasing the influence of borrower i's marginal cost on the optimal group price. Similarly, a higher internalization parameter υ_{kj} implies that bank k accounts more heavily for the profit displacement experienced by bank j. Consequently, the costs associated with bank j's most price-sensitive borrowers become increasingly relevant for bank k's pricing decision. The resulting price vector, given the profit internalization matrix $\mathbf{\Upsilon}_{gmt}$ and marginal costs $\mathbf{mc}_{imt}$, constitutes the Nash equilibrium for borrower group g in market m at time t.

The pricing equation nests several benchmarks from the literature. When no bank internalizes competitors' profits (i.e., $\upsilon_{kj} = 0$), the model reduces to standard Bertrand-Nash competition. In this case, each bank prices based on its own (weighted average) marginal cost plus a markup determined by borrower-group demand elasticity, a single-product version of Benetton (2021) and Benetton et al. (2025). The only adjustment to marginal cost comes from default

risk, wherein the bank discounts expected revenue by $(1 - d_{ikmt})$, the standard risk adjustment in models of loan pricing.

When banks do internalize competitors' profits ($\upsilon_{kj} > 0$), two additional features emerge. First, the effective marginal cost becomes a weighted average of own and competitors' costs, with weights reflecting substitution patterns and the degree of internalization. Second, the default risk adjustment now incorporates competitors' default rates as well since bank k cares that borrowers switching to bank j may default there. In the limiting case where default rates are zero and marginal costs are constant within borrower group, the pricing equation (Equation 6) reduces to the internalization pricing equation of Miller and Weinberg (2017).

2.2.1 Marginal Cost Assumption

Empirically, for each candidate model M and its internalization matrix $\boldsymbol{\Upsilon}^M$, we obtain model-consistent marginal costs $\mathbf{mc}_{gmt}(\boldsymbol{\Upsilon}^M)$ by inverting the first-order conditions and using prices, estimated default rates $\widehat{\mathbf{d}}_{imt}$, and estimated demand parameters. This inversion exploits the assumption that banks face a constant marginal cost within group–bank–market–year cells. In other words, the cost of supplying an additional loan to a given group does not vary with loan volume.

We validate this assumption in several ways. First, Internet Appendix A provides institutional evidence that banks were not funding-constrained around the SOLCA tax: Figure A4 shows that key liquidity ratios (liquid assets relative to short-term deposits and total assets) remained stable, with no visible deterioration around October 2014 or in the post-tax period.

Second, we specify marginal costs as flexibly as possible. We allow them to be (1) group-specific, capturing heterogeneous monitoring costs across firm types; (2) bank-specific, reflecting differences in efficiency; and (3) market-specific, permitting geographical variation related to, for example, local branch density. Furthermore, although marginal costs are constant within a given group, the composition of borrowers can still affect the bank's total cost function, allowing it to exhibit increasing, decreasing, or constant returns, depending on the selection patterns among borrowing firms, as in Cabral et al. (2018). All cost components also vary over time.

Third, we develop a marginal cost curvature test using group-bank-level cost estimates and a demand shifter. We fail to reject constant marginal costs under any assumed conduct model (Internet Appendix G).[22]

[22]The point estimates suggest mild dis-economies of scale, but these are economically small: moving from the 25th to the 75th percentile in loan size increases marginal cost by only 11% relative to average cost.

2.2.2 Accounting for Selection

We treat default risk at origination, d_{ikmt}, as predetermined at the pricing stage, meaning that banks form their predicted default probability based on borrower observables before setting prices. This differs from Crawford et al. (2018), who endogenize the default decision and correlate demand and default shocks, but is supported by the data in our setting. First, base default rates are low (see Table 3 and Internet Appendix A).[23] Second, the SOLCA tax shock has no statistically significant direct effect on default (Internet Appendix Table C2). Third, a test of correlations between residualized prices and default shows that while the relationship is statistically significant, the effects are economically negligible (Internet Appendix C). Together, this evidence indicates that adverse selection unobserved to the bank is not a first-order concern, and is broadly consistent with previous evidence that moral hazard has second-order effects in developing countries (Castellanos et al. (2025)).

While unobservable selection and moral hazard are second order in our setting, we do account for selection on observables. First, we allow marginal costs to vary flexibly at the group–bank–market–year level. Although both parties know this information, this captures the aggregate positive or negative selection induced by changes in interest rates, as in Cabral et al. (2018). Second, we allow for common-knowledge heterogeneity in default rates that depends on borrower characteristics, as in Benetton et al. (2025). Because demand for credit also depends on these same characteristics, our model captures the correlation between default risk and willingness to pay, i.e., selection patterns based on observables (Einav et al. (2021)).

2.3 Identification of the Internalization Matrix

As in prior literature (Bresnahan (1982); Berry and Haile (2014)), our model alone does not allow separate identification of the supply parameters without additional information. To understand why, suppose the econometrician has identified the demand and default parameters, either through standard estimation or via experimental measurement. Inverting Equation 6 then recovers marginal costs at the group–bank–market–year level, but conditional on a maintained conduct assumption Υ_{mt}. Different conduct assumptions yield different implied marginal costs.

However, the SOLCA tax is an observable quasi-experimental cost shifter. This additional variation allows us to separately identify marginal cost and Υ_{gmt}, conditional on demand and observed prices and quantities. The key is that the tax generates observed pass-through, which depends on conduct. Consider the pass-through $\text{PT}_{gmt}(\tau, \Upsilon^0_{gmt})$ induced by the vector of tax

[23] In Crawford et al. (2018), default rates are on average 6%, whereas in our full sample they are close to 0%.

shocks τ under true conduct Υ^0_{gmt}

$$\text{PT}_{gmt}(\tau, \Upsilon^0_{gmt}) \equiv \frac{\partial \tilde{\mathbf{r}}^e_{gmt}(\tau, \Upsilon^0_{gmt})}{\partial \tau} = \mathbb{1} + \frac{\partial \boldsymbol{\mu}^e_{gmt}(\tau, \Upsilon^0_{gmt})}{\partial \tau}, \tag{9}$$

where $\tilde{\mathbf{r}}^e_{gmt}(\tau, \Upsilon^0_{gmt})$ is the equilibrium tax-inclusive interest rate, $\boldsymbol{\mu}^e_{gmt}(\tau, \Upsilon^0_{gmt})$ is the equilibrium markup adjustment, and the bank's non-tax marginal costs of lending $\widetilde{\mathbf{mc}}_{gmt}(\Upsilon^0_{gmt})$ are unchanged by the tax. As long as pass-through is nonzero, i.e., $\frac{\partial \boldsymbol{\mu}^e_{gmt}}{\partial \tau} \neq -1$, then each pass-through measurement provides an additional moment condition for identifying Υ_{gmt}. With W such measurements (across groups g, markets m, and time t) and the $G \times B \times M \times T$ first-order conditions from Equation 6, we obtain an exactly-identified system: $G \times B \times M \times T$ equations pin down marginal costs and W pass-through moments pin down W conduct parameters.

While model identification holds, model estimation faces two practical limitations. First, empirical pass-through estimates, $\widehat{\text{PT}}_{gmt}$, are often noisy, making it infeasible to recover all elements of Υ^0_{gmt}. A natural simplification is to consider symmetric partial cartels C, where the internalization parameter is constant across all members ($\upsilon_{kj} = \upsilon_C$ if $k, j \in C$), and zero otherwise. Under this restriction, the econometrician can recover internalization parameters for up to W non-overlapping cartels.

Second, even when estimation is feasible, it remains subject to bias and misspecification. MacKay et al. (2014) show that pass-through estimates may be biased if cost shifters are correlated with unobserved demand or cost factors, which would bias the recovered υ_C. Moreover, if the assumed cartel memberships are incorrect, the model is misspecified. For both reasons, Magnolfi and Sullivan (2022) show that testing approaches (e.g., Backus et al. (2024); Duarte et al. (2024); Dearing et al. (2024)) are preferable to estimation.

2.4 Testing the Internalization Matrix with Selection

We adapt the conduct-testing logic of Backus et al. (2024), Duarte et al. (2024), and Dearing et al. (2024) to a setting with (1) borrower-level heterogeneity in demand and default, and (2) pricing and marginal costs recovered at the borrower-group level from a default risk-adjusted pricing equation. The core idea is to treat each candidate internalization matrix Υ^M as a competing model M of markups. Given demand and default estimates, each Υ^M implies a model-consistent series of marginal costs; we then ask whether these implied costs are orthogonal to the exogenous cost shifter (the SOLCA tax). Models that violate this restriction are rejected.

Model-implied marginal costs. Fix a candidate conduct model M with internalization matrix Υ^M. Given observed prices $\mathbf{r}_{gmt}$, estimated default rates $\widehat{d}_{imt}$, and estimated demand objects (so that $\boldsymbol{\Delta}_{imt}$ and $\boldsymbol{\Delta}^d_{imt}$ are known), the pricing equation (Equation 6) delivers the model-implied,"virtual" marginal costs

$$\widehat{\widetilde{\mathbf{mc}}}^{\,M}_{gmt} \equiv \mathbf{r}_{gmt} - \widehat{\boldsymbol{\mu}}_{gmt}(\Upsilon^M), \qquad \widehat{\boldsymbol{\mu}}_{gmt}(\Upsilon^M) = -\Big[\sum_{i\in g} \Upsilon^M \circ \widehat{\boldsymbol{\Delta}}^d_{imt}\Big]^{-1} \sum_{i\in g} \widehat{\mathbf{Q}}^d_{imt} \tag{10}$$

under our maintained assumption that marginal cost is constant within group-bank-market-year cells, $\widehat{\widetilde{mc}}^{\,M}_{gkmt}$ denotes the k-th element of $\widehat{\widetilde{\mathbf{mc}}}^{\,M}_{gmt}$.

Cost structure and exclusion restriction. Let the true (group-level) marginal cost satisfy

$$\widetilde{mc}^{\,0}_{gkmt} \;=\; h_0(w_{gkmt}) + \omega^0_{gkmt}, \tag{11}$$

where w_{gkmt} collects observed cost shifters (e.g., bank, market, and group characteristics) and ω^0_{gkmt} is an unobserved cost shock. Let z_{gkmt} denote an excluded instrument (in our application, the SOLCA tax interacted with a post-period indicator variable for private banks and potential cartel members). The maintained exclusion restriction is

$$\mathbb{E}\left[\omega^0_{gkmt}\, z_{gkmt} \mid w_{gkmt}\right] = 0. \tag{12}$$

For each conduct model M, define the model-specific residual

$$\omega^M_{gkmt} \;\equiv\; \widehat{\widetilde{mc}}^{\,M}_{gkmt} - h_M(w_{gkmt}), \tag{13}$$

where $h_M(\cdot)$ is the model-dependent best-fitting cost function within a flexible class (e.g., rich fixed effects and controls). A conduct model M is compatible with the data only if the corresponding residual satisfies the exclusion restriction

$$\mathbb{E}\left[\omega^M_{gkmt}\, z_{gkmt} \mid w_{gkmt}\right] = 0. \tag{14}$$

Intuitively, if Υ^M is incorrect, the implied markups in Equation 10 vary systematically with the tax in ways no admissible cost function $h_M(\cdot)$ can absorb, generating $\mathbb{E}[\omega^M z \mid w] \neq 0$.

Selection and the level of exogeneity required. Selection complicates testing because the group-level unobservable ω^0_{gkmt} is not a primitive borrower shock: it is an internalization-

and demand-weighted aggregation of borrower-level unobservables. To see this, consider a borrower-level decomposition under the true model,

$$\mathbf{mc}_{0imt} = \overline{mc}\left(L_{g(i)mt}, w_{g(i)mt}\right)\mathbb{1}_{K_{g(i)mt}} + \boldsymbol{\omega}_{0imt}, \tag{15}$$

where $\overline{mc}$ is a scalar function, $L_{g(i)mt} = \sum_{i\in g} L_{imt}$ and $w_{g(i)mt}$ capture equilibrium effects of loan size and demographics selection, and $\boldsymbol{\omega}_{0imt}$ is an idiosyncratic lending-cost component. Under our pricing inversion, the group-level unobservable shock entering Equation 11 is

$$\boldsymbol{\omega}_{0gmt} = \Big[\sum_{i\in g} \boldsymbol{\Upsilon}^0 \circ \boldsymbol{\Delta}^d_{imt}\Big]^{-1} \sum_{i\in g} [\boldsymbol{\Upsilon}^0 \circ \boldsymbol{\Delta}_{imt}]\boldsymbol{\omega}_{0imt}. \tag{16}$$

Because the weights depend on equilibrium demand derivatives that may themselves respond to the instrument, borrower-level mean independence $\mathbb{E}[\boldsymbol{\omega}_{0imt} z_t \mid w] = 0$ is generally insufficient to satisfy Equation 12. We therefore maintain the exclusion restriction directly by assuming group-bank-market-year-level mean independence.

Notice that in markets with selection, traditional instruments may violate the exclusion restriction, as ω_{0gmt} will likely be correlated with such instruments. For instance, demographic-based instruments such as firm growth and characteristics-based instruments like branch access do not satisfy mean independence, as demographics co-move with idiosyncratic cost shocks while characteristics are likely responses to changes in underlying cost shocks. For that reason, we consider tax-based instruments.

Instrument relevance. Conduct testing requires that the excluded instrument shift equilibrium pricing differently across candidate models. In our context, pass-through, as defined in Equation 9, must be non-zero and heterogeneous across models. In Section 3, we document that pass-through is statistically different from zero and varies across regions and banks, suggesting that the tax provides informative variation for distinguishing among candidate conduct structures $\boldsymbol{\Upsilon}^M$.

Constant marginal costs. A key maintained approximation is that marginal costs are locally constant with respect to loan volume at the group-bank-market-year level. If marginal cost curvature is economically important, then the tax can affect marginal costs endogenously through its impact on quantities and borrower composition (via $L_{g(i)mt}$ in Equation 15), violating the interpretation of Equation 12 as a pure exclusion restriction. We address this concern in two ways. First, we specify $h_M(w)$ flexibly to absorb observed sources of selection. Second, we

implement an explicit test for marginal cost curvature as a function of group-level $L_{g(i)mt}$ where we fail to reject constancy (Internet Appendix G).

Moral hazard. Instrument validity also requires that interest rates and taxes do not materially affect repayment behavior in ways not captured by observables. If higher rates induced higher default probabilities, then d_{imt} would respond endogenously to the tax, contaminating the cost inversion in Equation 10 and generating spurious correlation between z and ω^M. Reassuringly, default risk responds little to pricing after controlling for the borrower bins used for pricing. Moreover, the introduction of the SOLCA tax has no statistically-significant direct effect on overall default at the borrower-group level (Internet Appendix C).

3 The Empirical SOLCA Tax Pass-Through

In this section, we measure the pass-through of the SOLCA bank tax to contracted loan interest rates. These estimates serve two purposes for the structural analysis: they support the quasi-exogeneity of tax exposure and they establish instrument relevance for the non-nested conduct comparisons in Section 5. We then describe pooled pass-through estimates and show how pass-through varies with market-competitiveness proxies, which motivates the candidate conduct structures tested later.

3.1 Checking Pass-through Identification Assumptions

The conduct tests in Section 5 rely on the tax-based shifter z_{gkmt} satisfying two conditions: (1) it must generate informative variation in pricing (instrument relevance) and (2) be orthogonal to unobserved cost components (the exclusion restriction). This subsection validates both. The identifying assumption is that, absent the tax, contracted interest rates for the same borrower-bank pair would have followed parallel paths over time, conditional on borrower- and relationship-specific unobservables, contract terms, and predicted default risk. We probe this assumption by testing for pre-trends, confirming that loans from public banks that are exempt from the tax show no contemporaneous price response, and ruling out confounding shocks to demand or funding costs. If instead the observed price response reflected such concurrent shocks correlated with tax exposure, the exclusion restriction would fail (MacKay et al. (2014)). We also document that the tax generates a non-zero and persistent response in loan rates, establishing the relevance condition for non-nested comparisons (Dearing et al. (2024)).

We begin with event studies that transparently show the evolution of interest rates over time.

Consider the following specification for loan l contracted by firm i from bank k at time t:

$$r_{likt} = \sum_{j=-8}^{3} \delta_j 1\{t \in j\} + \beta_a ln(A_{likt}) + \beta_m ln(M_{likt}) + \alpha_i + \alpha_k + \eta DP_{likt} + \varepsilon_{likt}, \tag{17}$$

where r is the pre-tax interest rate, A is the loan amount, M is the maturity in years, α_i and α_k are firm and bank fixed effects, DP is the predicted default probability at origination, and ε captures time-varying unobservables.[24] Firm (bank) fixed effects control for time-invariant unobservable firm (bank) characteristics. In alternative specifications, we use firm-by-bank fixed effects to absorb stable, relationship-specific factors, e.g., match quality. This specification also controls for compositional change in borrower risk, as estimation is within already-active firm-bank pairs. Periods j are quarters relative to 2014 quarter 4, when the SOLCA tax took effect.

We control for loan maturity because it has a direct negative effect on contracted interest rates (Internet Appendix Table A1). Moreover, as we document below, the tax reduced both contracted maturity and loan amount. Excluding these controls would therefore bias δ_j upwards, *towards* complete pass-through. To prevent contamination from loans granted during the month the tax took effect, we drop all loans originated in October 2014. We normalize $\delta_{-2} = 0$, so that all coefficients are measured relative to two quarters before the tax was implemented. Standard errors are clustered at the bank-quarter level to account for within-bank correlation of pricing in a given quarter.

The coefficient δ_j measures the average percentage-point change in nominal interest rates on new loans in quarter j relative to two quarters before the tax. If $\widehat{\delta_j} < 0$, then pre-tax interest rates fell in response to the tax, indicating incomplete pass-through of the statutory borrower tax into tax-inclusive borrowing costs (i.e., banks partially offset the tax burden by lowering equilibrium rates).[25] If, instead, $\widehat{\delta_j} > 0$, there is more-than-complete pass-through, as borrowers pay both the tax and a higher interest rate. Lastly, if $\widehat{\delta_j} = 0$, pass-through is complete, i.e., borrowers pay the entire tax and banks do not adjust rates. If we assume a constant marginal cost, either incomplete or more-than-complete pass-through is evidence of imperfect competition in the commercial bank lending market.[26]

[24] We predict default at the loan level by regressing an indicator for the loan becoming 90 days or more past due on lagged, firm-level default predictors. See Internet Appendix C.3 for details.

[25] Although the statute requires the borrower to pay the tax, which is collected and remitted by the bank at origination, the economic incidence need not match the statutory incidence. To the extent banks lower interest rates, they bear some of the tax burden.

[26] Under constant demand curvature, incomplete pass-through implies log-concave demand, while more-than-complete pass-through implies log-convex demand.

Figure 1 presents the estimated coefficients from Equation 17, showing how nominal interest rates on private commercial loans (not inclusive of the tax) evolved around the tax's introduction. This specification allows us to test visually for pre-trends. Interest rates remained flat for eight quarters before the tax introduction. We cannot statistically distinguish pre-period coefficients from the normalized period ($j = -2$). Immediately after the tax took effect, nominal interest rates fell by around 0.2 percentage points, with a slight downward drift thereafter. If we assume a flat 0.5 percentage-point tax rate, this magnitude implies pass-through of approximately 60%, on average. That is, borrowers bear about 60% of the tax burden while banks absorb the remainder.[27] Results are similar when we replace separate bank and firm fixed effects with bank-by-firm pair fixed effects (right panel of Figure 1).[28] This specification reinforces that compositional changes in borrower risk do not drive our results, as the effects are identified within already-active lending relationships.

[Place Figure 1 here.]

One concern is that regulatory or macroeconomic changes concurrent with the tax could bias estimates toward incomplete pass-through. As a falsification test, we estimate Equation 17 on loans originated by government banks that were exempt from the SOLCA tax but were exposed to the same industry-wide shocks. General equilibrium effects mean the government sector does not work as a pure control, but limited substitution between the private commercial loan market and public lending minimizes spillovers in practice (Internet Appendix A). Government bank loans, therefore, serve as a reasonable check for other macro-shocks, despite differences in loan characteristics and borrower composition between private and public banks. We observe no evidence of anticipation or response to the SOLCA tax in Figure 2. Nominal interest rates on public loans fluctuate cyclically but are never significantly different from the $j = -2$ baseline, either before or after the tax goes into effect. This is true even though the coefficients are estimated more precisely than in our private loan sample.

[Place Figure 2 here.]

These event-study and falsification results establish two ingredients that are central for our conduct tests. First, they show that the SOLCA reform generates a persistent change in contracted rates for taxed private banks, implying a non-zero first stage when tax exposure is used

[27] Calculated as (0.5-0.2) / 0.5 = 0.6. This interpretation assumes the full 0.5% tax rate applies to all loans. In reality, the one-time charge on loans with maturities under one year is prorated by maturity. We address this below.

[28] Our granular data allow us to observe individual bank-firm relationships. Pair fixed effects absorb additional factors, such as firm-specific monitoring costs or pair-specific match quality.

as an excluded shifter. Second, the absence of a contemporaneous response for exempt public banks and the lack of pre-trends support the quasi-exogeneity of tax exposure (Equation 12) that underpins the orthogonality condition used to compare conduct models in Section 5.

We report additional robustness tests in Internet Appendix B. Our main analyses use a three-quarter post-period because, as the window lengthens, confounders increasingly threaten identification. To check we are capturing an equilibrium adjustment rather than a transitory price response, panels (a) and (b) of Internet Appendix Figure B1 extend the horizon to eight quarters after the tax and document that incomplete pass-through persists.

Internet Appendix Table B1 confirms incomplete pass-through using a difference-in-differences (DiD) design that compares treated loans (private banks, taxed post-reform) to control loans (public banks, not taxed) around the SOLCA tax. The coefficients remain stable relative to those obtained from the event-study with treated banks alone. Internet Appendix Figure B2 provides dynamic DiD coefficients, revealing no significant pre-trends between private and public loan prices and similar post-event dynamics, consistent with the null effects for public banks documented above.

We also test robustness to two contemporaneous shocks. First, global oil prices collapsed in mid-2014 (Internet Appendix Figure B3, panel (a)). Since Ecuador is an oil-dependent economy, this could affect interest rates. Our event-study estimates remain quantitatively similar and statistically significant after controlling for a cubic polynomial in oil prices (Internet Appendix Figure B3, panel (b)).

Second, the oil shock and contemporaneous bank regulation may have affected banks' marginal costs through changes in deposit rates (a measure of the cost of funds for banks). Average deposit rates did rise in 2014 (Internet Appendix Figure B4, panel (a)).[29] If deposit rate increases represent unobserved cost shocks concurrent with the tax, pass-through estimates would be biased upward, as the total cost shock (deposit rates plus loan tax) would exceed what the SOLCA tax alone can explain. Panel (b) of Internet Appendix Figure B4 shows that pass-through of the tax to interest-rate spreads (pre-tax nominal rates minus average deposit rates) is even more incomplete than our baseline estimates, though confidence intervals widen.

We further check that changes in aggregate credit volume are not driving results through non-constant returns to scale. Internet Appendix Figure B5 examines total commercial credit around when the tax goes into effect. Although point estimates are negative, we find no statistically significant change in aggregate credit through the first two quarters after the tax relative to the pre-period. Thus, at least for the early post-tax period, our estimates capture the direct

[29] We calculate bank-year deposit rates by weighting Central Bank-reported rates for different deposit maturities by each bank's product-level deposit shares.

effect of the tax on prices rather than indirect effects through changes in the marginal costs of banks.[30] Together, these tests support that contemporaneous tax exposure is not proxying for concurrent macroeconomic or funding-cost shocks, supporting the use of the SOLCA-based shifter as plausibly excluded in the conduct tests.

Finally, the theory of tax incidence under imperfect competition links price pass-through to market conduct (Weyl and Fabinger (2013); Pless and van Benthem (2019)), so we focus primarily on interest rate responses. However, maturity and loan amount are set in conjunction with interest rates and cannot be ignored. For example, Internet Appendix Table A1 documents robust negative correlations between both amount and maturity and nominal interest rates.

We therefore examine the tax's effect on other loan terms. Figure 3 reports event studies with loan maturity (left panel) and loan amount (right panel) as outcomes. The left panel shows that the maturity of new commercial loans decreased after the SOLCA tax was implemented, which is intuitive given the tax schedule's kink at the one-year maturity. The right panel shows that loan amounts also declined, reaching statistical significance by three quarters post-implementation. In contrast to the immediate and persistent effect on interest rates, these adjustments are gradual, consistent with interest rates being the primary margin of bank competition. Internet Appendix Figure B6 extends the horizon and reveals that, unlike interest rates, both amount and especially maturity revert toward pre-tax levels within eight quarters.

[Place Figure 3 here.]

Since the tax reduced both maturity and loan amount, excluding these variables would bias pass-through estimates upward toward complete pass-through. We therefore control for contemporaneous maturity and loan amount in all specifications. We include these controls semi-parametrically via bins rather than log-linearly, as this approach offers more conservative estimates by flexibly capturing non-linear relationships.

3.2 Estimating the Tax Pass-through Directly

The event study specification described by Equation 17 is useful because it allows us to test for any evidence of pre-trends in contract terms in anticipation of the introduction of the SOLCA tax and to examine the evolution of the response. However, because there is a kink in the

[30]To reconcile muted aggregate credit responses with relatively elastic demand for credit, notice that pass-through estimates from the specification with separate bank and firm fixed effects are lower than those with pair fixed effects, indicating that new borrowers experience more incomplete pass-through and thus smaller price increases.

tax percentage at a loan maturity of one year, we can only recover an imprecise average pass-through. We therefore directly measure the pass-through of the tax to the cost of borrowing. That is, we estimate how final, tax-inclusive prices change with respect to the amount of the tax for each loan. We estimate for loan l contracted by firm i from bank k at time t:

$$rTax_{likt} = \text{PT} \times tax_{likt} + \sum_{j=1}^{20} \beta_a^j 1\{A \in h\} + \sum_{j=1}^{20} \beta_m^j 1\{M \in z\} + \alpha_d DP_{likt} + \alpha_i + \alpha_k + \varepsilon_{likt}, \quad (18)$$

where $rTax$ is the tax-inclusive, annualized interest rate and tax is the loan's effective tax rate.[31] Following the structure of the SOLCA tax, for loans with a maturity of one year or longer tax is 0.5% after the reform and zero beforehand. For loans with less than a one-year maturity, tax is 0.5% $\times$ M, where M is the loan's maturity in years. Then $rTax$ is the nominal interest rate in percent plus tax—the tax-inclusive price of borrowing.[32] Control variables include flexible controls for the amount (A) and maturity (M) of the loan using 20 buckets; the loan maturity, M, with its 20 corresponding buckets; firm fixed effects α_i; bank fixed effects α_k; the predicted default probability DP; and time-varying unobservables captured by ε. As mentioned above, we control semi-parametrically for maturity and amount rather than log-linearly as it will offer more conservative estimates. The estimation window is from eight quarters before the introduction of the tax through three quarters afterward.

Column (1) of Table 4 reports the estimated direct pass-through, $\widehat{\text{PT}}$, of the tax to tax-inclusive interest rates on commercial loans granted by private banks. Hypothesis testing is conducted against the complete pass-through null hypothesis ($\text{PT}_o = 1$). If $\widehat{\text{PT}} < 1$ it indicates incomplete pass-through and $\widehat{\text{PT}} > 1$ corresponds to more-than-complete pass-through. We find that there is, on average, incomplete pass-through of the tax in aggregate. In particular, the borrower pays approximately 35% of the SOLCA bank tax on the average loan while the bank shoulders the rest by reducing the interest rate. Column (2) adds the probability of loan default as a control. The estimated coefficient remains statistically indistinguishable from that of column (1).

[Place Table 4 here.]

Columns (3) and (4) differ from columns (1) and (2) in that the estimation includes bank-

[31] Papers that run this type of empirical specification to recover pass-throughs include Atkin and Donaldson (2015); Pless and van Benthem (2019); Stolper (2024); and Genakos and Pagliero (2022).

[32] There is a slight abuse of notation since interest rates compound annually, but taxes are collected only once at the start of the credit. However, most loans in our sample have a term-to-maturity of less than one year, making this distinction minimal. Unreported results show nearly identical estimates when focusing on loans with a maturity of one year or less.

firm pair fixed effects instead of separate bank and firm fixed effects. Note that this specializes our analysis to established lending relationships with new loans both before and after the SOLCA tax was introduced. The pass-through remains incomplete, but the borrower now shoulders a higher proportion of the tax, roughly half rather than around a third of the tax burden. The point estimate is again statistically indistinguishable with and without including the probability of loan default as a control. Comparing the model specifications, a higher pass-through within relationships that are already established might indicate that these relationships have lower demand elasticity, all else equal, while firms that switch banks have more elastic demand (Weyl and Fabinger (2013); Ganapati et al. (2020)). Moreover, the fact that we find incomplete pass-through within pre-established relationships is also evidence that credit terms are not fixed long-term, and thus, a period-by-period model of pricing is accurate, even in long-term relationships.

As above, we perform robustness checks on the pass-through estimates controlling for potential time-varying confounders. In Internet Appendix Table B2, we run the same pass-through regressions as above, focusing on the pair fixed effect model and still find incomplete pass-through estimates between 0.4 to 0.7 under specifications controlling for oil or quarterly province GDP growth and in specifications that identify off within-period heterogeneity in the tax impact via year-quarter fixed effects, which exploit the tax differences across borrowers within the same quarter-year. These results again provide support that contemporaneous shocks do not drive the incomplete pass-through estimates.

3.3 Heterogeneity by Market Competitiveness

We now provide evidence that pass-through is indicative of differences in market power and conduct across markets. For this empirical exercise, we define a market as the city in which the headquarters of the firm is located. We test if observed pass-through from the SOLCA tax is higher (lower) where we observe more (less) competitive conditions. The hypothesized positive relationship between tax pass-through and competitive market structure is based on the insight from Weyl and Fabinger (2013) and others that pass-through is higher under Bertrand-Nash competition than under joint maximization if demand is log-concave.[33]

To explore heterogeneity in the estimated treatment effect by competition proxies, we con-

[33] When demand is log-concave, pass-throughs are incomplete, like in our setting.

sider the following model:

$$rTax_{likt} = \text{PT} \times tax_{likt} + \delta_h tax_{likt} \times X_{likt} + \sum_{j=1}^{20} \beta_a^j 1\{A \in h\} + \sum_{j=1}^{20} \beta_m^j 1\{M \in z\} + \alpha_d DP_{likt} + \alpha_{ik} + \varepsilon_{likt}, \tag{19}$$

where X_{likt} is some market or firm characteristic, such as number of lenders by firm, or number of lenders in the market, etc. Coefficient δ_h captures the heterogeneity in the treatment effect. All interacted variables are standardized such that the main effect is the pass-through for the average borrower. We use the same time windows as in the event-studies, namely, eight quarters before and three quarters after the policy.

Table 5 presents the results. In column (1), we study pass-through heterogeneity in terms of the availability of lenders in the market, where the variable # *Potential Lenders* is the maximum number of active lenders that have historically lent in the market up to October 2014. We find that markets with more potential lenders have pass-throughs that approach Bertrand-Nash, i.e., where the pass-through is closer to the competitive benchmark of full pass-through (PT = 1).

[Place Table 5 here.]

Turning to column (2), if the market is more concentrated, as indicated by a higher value of *HHI City*, the average yearly Herfindahl-Hirschman index (HHI) based on commercial lending share in the firm's city, then estimated pass-throughs are lower, i.e., further from the Bertrand-Nash benchmark of complete pass-through. Moreover, in column (3), we see that areas with higher *Multi−Market Contact*, or a higher number of other markets (provinces) where the same banks offer commercial loans, have pass-throughs in line with less competitive conduct.[34]

It is beyond the scope of this paper to fully demonstrate how coordination is sustained when pricing loans. However, in column (4), we present suggestive evidence that, along with the multi-market contact result in column (3), hints that frequent contact between banks may be part of the answer. The ASOBANCA banking association (BA) is a prominent bank lobbying group in Ecuador that organizes regular meetings and events between banks and whose website lists a primary purpose as promoting cooperation and communication between members. It is therefore a plausible mechanism for explicit or implicit coordination.[35] In column (4), we report that the SOLCA tax pass-through is less complete the higher the market share of BA

[34]This test is in the spirit of Ciliberto and Williams (2014) and Hatfield and Wallen (2023), which show that multi-market contact may facilitate tacit coordination and reduce competition.

[35]Internet Appendix E provides full institutional details and supporting evidence on lending rate patterns associated with BA membership.

members in the market. Motivated by this reduced-form pattern and the institutional context (Internet Appendix E), Section 5.1 treats BA membership as a candidate internalization group and formally tests it against a Bertrand-Nash benchmark.

Need for a model. Overall, results are consistent with the hypothesis that banks coordinate when setting prices implicitly and/or explicitly and that pass-through can capture heterogeneity in competition across markets. However, this is suggestive evidence that markets are not competitive. It may be that the form of competition is close to Bertrand-Nash yet markets differ widely in their demand determinants. For example, more concentrated markets may be smaller or in distant markets in which firms' investment needs are scant, either of which would affect the shape and curvature of the demand for capital. While the bank-firm pair fixed effects and controls for contract terms may capture some of this cross-market heterogeneity in demand, it might be insufficient if demand curves are non-linear. This is why a model is needed to flexibly control for demand and test whether the data departs from a Bertrand-Nash benchmark.

4 Estimating Demand

This section describes our estimation strategy for the demand parameters in Section 2, presents and interprets the resulting estimates, and assesses model fit.

4.1 Estimation Strategy

We estimate a discrete-continuous demand model using maximum likelihood, building on the random-coefficients discrete-choice literature (Berry et al. (1995); Conlon and Gortmaker (2025); Grieco et al. (2025)) and extending it to incorporate a continuous loan-size equation following Crawford et al. (2018).

Defining market m. While in the reduced-form section we explored heterogeneity at the city level, for estimation we define markets as regions, in order to reduce estimation noise. In particular, we consider the provinces Azuay, Guayas, and Pichincha, which have the most economic activity, as regions. We aggregate across provinces for the remaining coastal (Costa) and interior (Sierra/Oriente) regions. Demand estimation is conducted separately by region, so demand coefficients vary by region.

Defining group g and prices r. Estimation requires several preliminary steps. First, we define borrower groups g based on the interaction of four characteristics: (1) credit purpose (general loan vs. other); (2) segment (corporate, enterprise, SME, or other commercial); (3) risk (risky if risk weight < B, non-risky otherwise); and (4) maturity (four buckets).[36]

Second, because we observe only granted loans while the demand model requires a menu of prices from all banks to all potential borrowers, we must define choice sets and impute prices. We define each group's choice set as all active banks ever serving that group in the market and compute prices as group-level averages. Thus, a firm i in group g receives offers $r_{g(i)kmt}$ from all active banks k in the region m at time t. Details appear in Internet Appendix C.

Outside option. To allow for the opportunity to not borrow, we include a set of non-borrowers within each group. However, given differences in characteristics between borrowers and non-borrowers (Table 3), we create the sample of non-borrowers using a matching approach, as in Crawford et al. (2018), to find with replacement the nearest neighbor for each borrower. Internet Appendix C provides the details.

Subsampling. Due to computation constraints, we sample borrowers for estimation and counterfactuals. We use the full sample of borrowers to construct choice sets and group-level prices, then draw up to 10,000 borrower-years for estimation, retaining the full choice set for each.

Obtaining the likelihood function. With micro-data, we estimate the discrete-continuous model via maximum-likelihood, extending a standard mixed logit framework. The empirical discrete-choice Equation 1 has the following simplifications and assumptions:

$$\Pi^s_{ikmt}(\tilde{\delta}, \alpha, \beta, \sigma_\alpha, \sigma_\psi) = \tilde{\delta}^s_{g(i)kmt} + (\alpha_1^{s,x} X_{it} + \alpha_2^{s,x} X_{ikmt} + \tilde{\alpha}^s_{it}) r_{g(i)kmt} + \beta^s_1 X_{it} + \beta^s_2 X_{ikmt} + \psi^s_{it} + \varepsilon^s_{ikmt}, \tag{20}$$

where the group-bank-market-time fixed effects absorb the mean price response: $\tilde{\delta}^s_{g(i)kmt} = \delta^s_{g(i)kmt} + \alpha^s r_{g(i)kmt}$. We assume $\tilde{\alpha}^s_{it} \sim N(0, \sigma^2_\alpha)$, $\psi^s_{it} \sim N(0, \sigma^2_\psi)$, and $\varepsilon^s_{ikmt} \sim T1EV$.

Assuming the taste shock ε^s_{ikmt} is distributed i.i.d. Type-I Extreme Value, the conditional

[36] Segments are defined by the Central Bank of Ecuador based on the previous year's sales and thresholds vary over time. For 2014, the corporate threshold is > \$1 million; enterprise \$200,000-\$1 million; SMEs < \$200,000; other commercial includes firms not assigned a segment but classified as commercial credit. Credit purpose comes directly from the registry: general loans include working capital and term loans; other includes letters of credit, titles, credit cards, and other commercial instruments.

probability that firm i chooses bank k, conditional on the random coefficients, is

$$s_{ikmt}(\tilde{\delta}, \alpha, \beta, \sigma_\alpha, \sigma_\psi) = \frac{\exp\left(\Pi^s_{ikmt}(\tilde{\delta}, \alpha, \beta, \sigma_\alpha, \sigma_\psi)\right)}{1 + \sum_j \exp\left(\Pi^s_{ijmt}(\tilde{\delta}, \alpha, \beta, \sigma_\alpha, \sigma_\psi)\right)}, \tag{21}$$

where we normalize the indirect discrete-choice utility from not borrowing to zero. The unconditional probability integrates over the distribution of random coefficients:

$$S_{ikmt}(\tilde{\delta}, \alpha, \beta) = \int \int s_{ikmt}(\tilde{\delta}, \alpha, \beta, \sigma_\alpha, \sigma_\psi) dF(\psi_i) dF(\alpha_i). \tag{22}$$

Conditional on choosing bank k, the loan demand function from Equation 2 becomes:

$$\ln(L_{ikmt}) = \tilde{\delta}^l_{g(i)kmt} + (\alpha^{l,x}_1 X_{it} + \alpha^{l,x}_2 X_{ikmt}) r_{g(i)kmt} + \beta^l_1 X_{it} + \beta^l_2 X_{ikmt} + \varepsilon^l_{ikmt}, \tag{23}$$

where $\tilde{\delta}^l_{g(i)kmt} = \delta^l_{g(i)kmt} + \alpha^l r_{g(i)kmt}$ and $\varepsilon^l_{ikmt} \sim N(0, (\sigma^l_\varepsilon)^2)$. Under the assumption of normality for ε^l_{ikmt}, the conditional density of log loan size is:

$$f(ln(L_{ikmt})|k, k \neq 0) = \tag{24}$$
$$\frac{1}{\sqrt{2\pi(\sigma^l_\varepsilon)^2}} \times \exp\left[- \frac{\left(\ln(L_{ikmt}) - \tilde{\delta}^l_{g(i)kmt} - (\alpha^{l,x}_1 X_{it} + \alpha^{l,x}_2 X_{ikmt}) r_{g(i)kmt} - \beta^l_1 X_{it} - \beta^l_2 X_{ikmt}\right)^2}{2(\sigma^l_\varepsilon)^2} \right].$$

The joint log likelihood that firm i borrows amount L_{ik} from bank k is:

$$\ln(\mathcal{L}) = \sum_{t=0}^{T} \sum_{m=0}^{M} \sum_{k=0}^{K_m} \sum_{i=0}^{I_m} \mathbb{1}_{ikmt} \Big[\ln(S_{ikmt}) + \ln(f(\ln(L_{ikmt}) \mid k, k \neq 0)) \Big], \tag{25}$$

where $\mathbb{1}_{ikmt}$ equals one if borrower i chooses the loan offered by bank k, and zero otherwise. This likelihood function accounts for the simultaneity inherent in the discrete-continuous choice, where the firm jointly picks a bank and a loan size.

We implement this maximum likelihood procedure using a two-step approach (Grieco et al. (2025)).[37] In the first step, we estimate the group-bank-market-time constants $\tilde{\delta}^s_{gkmt}$ and $\tilde{\delta}^l_{gkmt}$; the coefficients $\alpha^{s,x}$, $\alpha^{l,x}$, β^s and β^l; and the variance parameters σ^2_ψ (discrete choice constant), σ^2_α (price), and $(\sigma^l_\varepsilon)^2$ (continuous choice). We initialize the optimization using the estimates from a simple logit model for the discrete choice and an OLS model for the continuous choice,

[37] While joint estimation of both steps offers efficiency gains (Grieco et al. (2025)), our likelihood is non-standard due to the continuous choice component.

then proceed via gradient-based search.

In the second step, we recover the mean price coefficients α^s and α^l from the estimated fixed effects ($\tilde{\delta}^s_{gkmt}$ and $\tilde{\delta}^l_{gkmt}$) via instrumental variables to address potential endogeneity of prices and measurement error from price imputation, pooling all markets together. For each equation e (discrete choice s and continuous choice l), we estimate

$$\widetilde{\tilde{\delta}}^e_{gkmt} = \alpha^e r_{gkmt} + \delta^e_k + \delta^e_t + \epsilon^e_{gkmt}, \tag{26}$$

where δ^e_k and δ^e_t are bank and year fixed effects. We instrument group-bank-market-time prices r_{gkmt} using: (1) prices of commercial loans in other markets; (2) delinquency rates in microcredit, mortgages, and consumer loans; and (3) changes in return on capital. The key identifying assumption is that these cost-based and Hausman-style instruments capture bank-level variation in marginal costs that is orthogonal to borrower-level demand shocks. As the specification includes year fixed effects, aggregate demand shocks are captured and the variation in the Hausman-style instrument is bank specific. Moreover, the inclusion of bank fixed effects eliminates level differences cross-sectionally.

4.2 Estimated Demand Parameters

Table 6 reports the demand parameter estimates. The parameters are estimated separately by region, and we report the mean and standard deviation of the point estimates across regions. For the price coefficients, we report the instrumental variable estimates pooling all regions.

The instrumental variable strategy requires that the instruments be relevant (correlated with prices) and satisfy the exclusion restriction (affect demand only through prices). We estimate statistically significant negative price coefficients for both margins: -0.599 for the discrete choice and -0.488 for the continuous choice. The Cragg-Donald Wald F-statistic is 11.08, indicating adequate instrument strength. While the exclusion restriction cannot be directly tested, we argue that these instruments respond to common bank-level cost factors and do not affect individual loan demand except through interest rates, as the Sargan-Hansen overidentification test fails to reject the null hypothesis of instrument exogeneity, with p-values of 0.77 and 0.73 for the discrete and continuous equations, respectively.

[Place Table 6 here.]

The remaining demand parameters in Table 6 have expected signs. Older firms are less likely to borrow, but conditional on borrowing, they take larger loans. Larger firms (measured by assets) are more likely to borrow and borrow more intensively, while firms with higher

revenue (a proxy for cash on hand) are less likely to borrow and borrow smaller amounts. We find other inputs (wages and expenditures) are complements to credit in the discrete choice, while intermediates are complements and labor is a substitute in loan intensity. More indebted firms are less likely to borrow and take smaller loans. Borrowers are more likely to choose banks the longer their lending relationship, and they borrow more from these primary lenders.

The interactions between firm characteristics and price show some heterogeneity. While specific coefficients vary by region, allowing for this flexibly is important for the conduct tests. Lastly, we find significant variance in the random coefficients on price and the constant, which we account for in the supply-side analysis. Regional estimates appear in Internet Appendix Table D1 and display similar patterns.

4.3 Estimates of Demand Elasticities

Internet Appendix F derives the elasticity formulas for the discrete-continuous model. Table 7 reports estimated own- and cross-demand elasticities. *Continuous* denotes the intensive margin (loan size) and *Discrete* denotes the extensive-margin (bank choice) elasticity. A one percent increase in interest rates reduces loan size by 3.45% on average (median: 3.39%) and market share by 4.76% (median: 5.01%).[38]

[Place Table 7 here.]

Total elasticity sums the intensive and extensive margins. Demand heterogeneity across borrowers is substantial: mean total elasticity is −8.21 (median: −8.25), with some borrowers reaching −16.53. That our model is flexible enough to capture this heterogeneity is critical, as it may explain variation in pass-throughs (Miravete et al. (2023)). Internet Appendix Table F1 confirms this by showing that simulated Bertrand-Nash pass-throughs range from well below to above one across borrower groups, so the demand specification does not mechanically impose incomplete or complete pass-through. Finally, *Cross* elasticity measures substitution to competing banks. We find that a one percent rate increase raises competitors' market shares by 0.62% on average (median: 0.12%).

We validate these structural elasticities against reduced-form IV estimates in Internet Appendix Figure F1. The two approaches align well. Moreover, Internet Appendix Figure F2 shows that elasticity heterogeneity by relationship age and firm size matches findings from Allen et al. (2019) on switching costs: older relationships are less price elastic and smaller firms are more price elastic.

[38] These estimates are slightly more elastic than those from Crawford et al. (2018) and Ioannidou et al. (2022), but similar in magnitude to those from Benetton (2021) and Benetton et al. (2025).

4.4 Model Fit

Table 8 reports descriptive statistics on model fit for market shares (discrete choice), loan size (continuous choice), prices, and default rates.[39] The model fits mean values well, with a perfect fit for market shares, loan use, and default rates. Our model under-predicts prices by a small margin. As expected, the model predicts less variation than in the data. Internet Appendix Section Appendix F.3 presents binscatter plots showing a good fit across the distribution.

[Place Table 8 here.]

5 Supply-Side Internalization Testing Framework

Having estimated demand (Section 4), we evaluate alternative models of lender competition using the non-nested conduct-testing framework developed in Backus et al. (2024); Duarte et al. (2024); Dearing et al. (2024) and adapted to our internalization-matrix setting following Miller and Weinberg (2017).

We introduce two modifications tailored to our setting. First, we incorporate price discrimination and default risk by recovering model-implied markups and implied (risk-adjusted) marginal costs at the borrower-group level and testing whether pricing is consistent with independent Bertrand-Nash competition or with partial profit internalization. Second, unlike settings where demand shifters can serve as instruments, selection and relationship-specific frictions make many demand-side instruments suspect in our context because they may also shift marginal costs, violating the exclusion restriction. We therefore rely on the SOLCA tax as an excluded markup shifter and implement the non-nested comparisons by asking which candidate internalization matrix generates implied marginal costs most consistent with the orthogonality condition in Section 2.4. The demand system is flexible enough that Bertrand-Nash alone generates pass-through both above and below one across borrower groups (Internet Appendix Table F1), suggesting any rejection of Bertrand-Nash in the tests below reflects supply-side pricing behavior rather than restrictive demand curvature.

Suppose equilibrium play under the true conduct model generates observations for borrower group g borrowing from bank k in market m at time t, with pricing given by Equation 6. Given any candidate conduct model M with internalization matrix Υ^M, we compute model-implied marginal costs net-of default-risk from Equation 10 and compare them across competing symmetric cartel models, where $\upsilon_{kj} = \kappa$ for all $k, j \in C$ in alleged cartel C. For instance,

[39] Internet Appendix C.3 discusses our procedure for estimating default rates.

Bertrand-Nash ($\kappa = 0$), joint maximization ($\kappa = 1$ for all k, j), and intermediate degrees of internalization ($0 < \kappa < 1$) applied to alternative cartel membership definitions (BA, Top-4, and Industry-Wide) to assess which best rationalizes observed pricing.

Instruments. For identification and testing, we use excluded instruments z_{gkmt} (Berry and Haile (2014)) that satisfy the exclusion restriction for the unobserved cost shock under the true conduct, $\mathbb{E}[\omega^0_{gkmt} z_{gkmt} \mid w_{gkmt}] = 0$ (Equation 12). All instruments come from the SOLCA tax. We use the 2013–2016 window around the reform and exclude firm-years that borrowed in the partially-treated fourth quarter of 2014. As mentioned above, the tax applies only to private lenders and its liability scales with maturity up to one year, so the reform generates variation in tax exposure across banks and borrower groups. We therefore measure exposure at the borrower-group × bank × market-year level by the effective tax rate

$$\tau_{gkmt} = \text{Post}_t \times 0.5 \times \min\left(\overline{\text{mat}}_{gkmt},\ 1\right) \times (1 - \text{Public}_k), \tag{27}$$

where $\overline{\text{mat}}_{gkmt}$ is the average annualized maturity of the cell's loans, Post_t indicates the post-reform years, and Public_k indicates state-owned banks, which are exempt. The baseline instrument vector contains Post_t, $\text{Post}_t \times \text{Public}_k$, τ_{gkmt}, and τ^2_{gkmt}; the quadratic term allows the pricing response to be non-linear in exposure. Because the models differ in which banks internalize one another's profits, each comparison adds the interaction of Post_t with membership in every candidate cartel involved in that comparison (the top-four banks or the Banking Association). This lets the instruments separate the implied markups of cartel members and non-members. Comparisons therefore use four to six instruments, and the model confidence set uses the six-instrument set containing both membership interactions.

RV test. Consider two candidate conduct models, M^1 and M^2. We compare them using the Rivers–Vuong (RV) test for conduct of Duarte et al. (2024), adapted to our risk-adjusted pricing equation. The null hypothesis is that both models fit the data equally well; the alternative is that one model fits better.

Cost shocks. For each model M, Equation 10 delivers the implied marginal costs $\widehat{\widetilde{mc}}^{M}_{gkmt}$, which are net of default risk. Moreover, following Equation 11, we separate implied marginal costs into a component explained by observed cost shifters and an unobserved cost shock. We specify $h_M(w_{gkmt}) = w'_{gkmt}\phi_M$, where w_{gkmt} contains average borrower-group characteristics (relationship age, firm age, and log assets, debt, revenue, capital, wages, and expenditures)

together with bank and region fixed effects. For each model we estimate ϕ_M by least squares and recover the model-implied cost shock

$$\widehat{\omega}^M_{gkmt} = \widehat{\widetilde{mc}}^M_{gkmt} - w'_{gkmt}\widehat{\phi}_M, \tag{28}$$

the empirical counterpart of Equation 13. The test is conducted on these cost shocks: model M is consistent with the data only if $\widehat{\omega}^M_{gkmt}$ is orthogonal to the excluded instruments conditional on the observed cost shifters (Equation 14).

Test statistic. Model M's lack of fit is the GMM objective formed from the sample analogue of Equation 14,

$$Q(M) = \mathcal{G}'_M \, \Lambda \, \mathcal{G}_M, \qquad \mathcal{G}_M = \frac{1}{n}\sum_{gkmt} \tilde{z}_{gkmt}\, \widehat{\omega}^M_{gkmt}, \qquad \Lambda = \Big(\frac{1}{n}\sum_{gkmt} \tilde{z}_{gkmt}\tilde{z}'_{gkmt}\Big)^{-1}, \tag{29}$$

where $\tilde{z}_{gkmt}$ denotes the instruments after the observed cost shifters w_{gkmt} have been partialled out, the same projection that produces $\widehat{\omega}^M_{gkmt}$, and n is the number of borrower-group $\times$ bank $\times$ region-year observations. A model whose implied cost shocks co-move with the tax exposure, after conditioning on observed cost shifters, has a larger $Q(M)$. The RV statistic is

$$T^{RV} = \frac{\sqrt{n}\,(Q(M^1) - Q(M^2))}{\widehat{\sigma}_{RV}}, \tag{30}$$

where $\widehat{\sigma}_{RV}$ is the estimated asymptotic standard deviation of $\sqrt{n}\,(Q(M^1) - Q(M^2))$. Under the null, T^{RV} is asymptotically standard normal, and a positive value favors M^2.

Instrument strength and multiple models. The RV test is informative only if the instruments move the two models' implied cost shocks differently, that is, if they separate the models' markups. When they are weak for testing, the test can have poor size and little power (Duarte et al., 2024). We therefore report, for each comparison, the effective F-statistic of Duarte et al. (2024), the critical value it must exceed for 95 percent best-case power, and ρ^2, the degeneracy measure of Duarte et al. (2024), which together with the number of instruments indexes the critical value. Because pairwise comparisons can be inconclusive when several models remain plausible, we also compute the model confidence set of Hansen et al. (2011), as adapted to conduct testing by Duarte et al. (2024), jointly over all sixteen candidate models using the full instrument set. The procedure repeatedly tests whether the remaining models fit equally well and removes the worst-fitting model when that hypothesis is rejected. Each model's p-value is

the significance level at which it is eliminated.[40]

5.1 Testing Results

We first test whether the data reject Bertrand-Nash competition ($\upsilon_{kj} = 0$ for all k, j).

Table 9 reports non-nested tests of Bertrand-Nash as the null model against cartel alternatives that vary along two dimensions: cartel membership and internalization degree (κ). A positive T^{RV} (column 3) indicates that the cartel model is preferred to Bertrand-Nash. Column 4 reports the effective F of Duarte et al. (2024) and column 5 the critical value it must exceed for the comparison to have adequate power.

[Place Table 9 here.]

Panel A tests Bertrand-Nash against the alleged Banking Association (BA) cartel, with $\upsilon_{kj} = \kappa$ for all pairs where both k and j belong to BA, and κ ranging from 0.2 (weak internalization) to 1.0 (joint maximization among BA members). BA has been subject to antitrust investigation (Internet Appendix E), and BA member market share is a significant driver of pass-through heterogeneity (Section 3.3). Panel B tests against a Top-4 cartel, with $\upsilon_{kj} = \kappa$ for k, j among the four largest banks by assets, all BA members. Panel C tests against an Industry-Wide cartel ($\upsilon_{kj} = \kappa$ for all k, j), full joint maximization. $\kappa = 1$ corresponds to full joint maximization across all banks.

The data reject Bertrand-Nash against every candidate cartel at every degree of internalization. Across the fifteen comparisons, T^{RV} ranges from 3.03 to 5.21, rejecting at the one percent level throughout, and the effective F ranges from 14.5 to 24.7 against critical values between 5.7 and 9.4. Every comparison therefore clears the threshold for best-case power of 95 percent. Bertrand-Nash cannot account for how lenders repriced loans after the tax without attributing part of that response to unobserved cost shocks, which the exclusion restriction rules out.

While we can confidently reject Bertrand-Nash at any degree of internalization, tests are not powered to distinguish between internalization levels within the candidate cartels. Internet Appendix Table I1 tests partial internalization ($\kappa < 1$) as the null against joint maximization ($\kappa = 1$) as the alternative, holding cartel membership fixed. The data do not identify the degree of internalization within any candidate cartel. Ten of the twelve comparisons fall below their

[40] We implement the tests with the `pyRVtest` package of Duarte et al. (2024). We supply the observed group-level prices r_{gkmt} and each model's risk-adjusted markups $\widehat{\mu}_{gkmt}(\Upsilon^M)$, so the package works with the risk-adjusted marginal costs in Equation 10, and we specify w_{gkmt} as the cost formulation. Because demand and default are estimated outside `pyRVtest`, the reported standard errors treat the estimated markups as given and do not adjust for first-step estimation error.

critical value, and the two that clear it both involve the industry-wide cartel and are inconclusive.

Next, Internet Appendix Table I2 compares cartel membership definitions at matched κ. All comparisons at $\kappa \leq 0.6$ clear their critical value and none of the six at $\kappa \geq 0.8$ does, so membership is identified at weak internalization. Every powered comparison favors the narrower coalition. The industry-wide cartel is rejected against both alternatives at $\kappa = 0.2$, 0.4 and 0.6, and the banking association is in turn rejected against the top-four cartel at those same values.

Finally, Table 10 reports model confidence set p-values across all candidate conduct models. The procedure is a single sequential elimination: it asks whether the models still under consideration fit equally well, drops the worst when that is rejected, and repeats until it can no longer reject. The reported p-value is the significance level at which that model is eliminated, revealing that Bertrand Nash conduct is the least consistent with the data. In addition, every industry-wide specification is excluded at the 10 percent level. Instead, the retained set at the 10 percent level is the top-four and banking-association cartels at every degree of internalization.

[Place Table 10 here.]

Although our goal is not to pinpoint the exact composition of a banking cartel, the testing results support two conclusions. First, Bertrand-Nash is rejected against every candidate cartel at every degree of internalization, and this is the comparison where the instruments are strongest. Second, the data do not identify which coalition is coordinating or how strongly, beyond ruling out an industry-wide arrangement and favoring the narrower coalitions at weak internalization. Taken together, the usual assumption of Bertrand-Nash competition is a bad approximation for conduct in this setting.

5.2 Importance of Conduct Assumption for Supply-Side Parameters

Having established that Bertrand-Nash is inconsistent with the data, we now examine how the conduct assumption affects estimated supply-side parameters. Table 11 reports marginal costs, markups, and Lerner indices under alternative conduct models.

[Place Table 11 here.]

Panel A reports bank-borrower group estimates under Bertrand-Nash. Average (median) marginal costs of 5.95 (6.77) percentage points per dollar lent reflect costs like funding, monitoring, and screening. The corresponding average (median) markup is 2.98 (3.21) percentage

points, yielding Lerner Indices of 0.5 (0.44). Even under differentiated Bertrand-Nash competition, markups are substantial.

We next examine estimates under alternative conduct assumptions, focusing on joint maximization ($\kappa = 1$) within each cartel membership definition. For the alleged BA cartel (Panel B), implied marginal costs are lower than under Bertrand-Nash at 4.96 percentage points on average (median: 5.94), or 17% (12%) below the Bertrand-Nash estimates. Observed prices can thus be rationalized with lower costs under this conduct model. Implied markups are correspondingly higher, with mean (median) of 3.97 (4.09) percentage points, and Lerner Indices of 0.68 (0.52).

For the alleged Top-4 cartel (Panel C), estimates lie between Bertrand-Nash and the alleged BA cartel: marginal costs of 5.76 (6.60), markups of 3.16 (3.44), and Lerner indices of 0.52 (0.45). For the alleged Industry-Wide cartel (Panel D), marginal costs are 5.59 (5.82), markups are 3.33 (4.12), and it has the highest Lerner indices at 0.75 (0.55).

As external validation, our model-free proxy for the marginal cost of funds for the average bank using deposit rates (approximately 4.7%, Table 3) lies furthest from the marginal costs implied by Bertrand-Nash and closer to those implied by every cartel specification. This untargeted moment supports our finding that assuming Bertrand-Nash competition overestimates lender marginal costs.

The differences in Lerner Indices across conduct assumptions may help explain why estimated markups in the structural banking literature are often modest.[41] Assuming competitive conduct when banks coordinate misattributes coordination rents to costs rather than markups.

6 Tax Incidence, Tax Revenue, and Conduct

Financial taxes and levies are pervasive. Taxes on loans exist across South America (Argentina, Brazil, Peru, Colombia, Venezuela, Bolivia, Ecuador), Africa (Egypt, South Africa), Europe (Hungary, Spain, Sweden), and Asia (Malaysia). Stamp duties on mortgage loans are widely used, including in the United Kingdom, India, and Spain, and bank levies are common, especially in Europe. Moreover, while generally considered distortionary (Restrepo (2019)), financial transaction taxes are frequently proposed, including in the Inclusive Prosperity Act introduced by Senator Bernie Sanders in 2019 and COM/2013/71 proposed by the European

[41] For instance, Benetton (2021) finds markups of 18% of the average interest rate, while Crawford et al. (2018) reports markups of only 5%.

Commission in 2013.[42] However, the literature has not reached consensus on who bears the burden of financial taxes. Theoretically, the answer depends on demand and supply parameters, including supply conduct. Nevertheless, empirical work on such taxes implicitly or explicitly assumes Bertrand-Nash competition, i.e., zero internalization of competitors' profits.

We fill this gap by using our model to study how bank conduct affects the tax incidence, or who bears the burden of financial taxes. We also simulate the efficiency cost of the tax, as measured by its excess burden (the deadweight loss per dollar of tax revenue raised), and examine how both incidence and efficiency depend on the market structure of commercial lending.

To do so, we compare outcomes under Bertrand-Nash, the alleged BA cartel with $\kappa = 0.2$ and $\kappa = 1$, and the alleged Top-4 cartel with $\kappa = 0.2$ and $\kappa = 1$. For each conduct model, we first recover marginal costs consistent with observed prices and quantities, then simulate the introduction of a uniform 0.5% unit tax and recalculate Nash equilibrium prices. Using the simulated pre-tax and post-tax equilibria, we calculate borrower surplus (CS), bank surplus (PS), and tax revenue, aggregating at the borrower level. Formulas are reported in Internet Appendix H. We perform this exercise using data from before the tax was introduced, in the spirit of ex-ante policy evaluation.

Table 12 presents the results for each conduct regime, and Internet Appendix Table H1 reports each specification's difference from Bertrand-Nash, computed within bootstrap draw. The lower tax pass-through under profit internalization (Internet Appendix Table F1) and the incidence results of Weyl and Fabinger (2013) predict the sign of these differences: borrowers lose less, banks lose more, tax revenue is higher, and incidence, the borrower share of revenue, and the net excess burden are lower. To aid one-sided testing related to these differences, we report an 80% confidence interval that excludes zero on the predicted side corresponding to a one-sided test at the 10% level. As such, both tables report parametric bootstrap standard errors and 80% bias-corrected bootstrap intervals (Internet Appendix H).

Panel A (Table 12) reports median borrower-level changes in CS, PS and tax revenue. At weak internalization ($\kappa = 0.2$), the Panel A outcomes are economically indistinguishable from Bertrand-Nash for both cartels for the majority of outcomes, so we focus on full internalization ($\kappa = 1$). Under full internalization, both cartels shift the burden of the tax from borrowers to banks. The decline in borrower surplus is 4% smaller under the Top-4 cartel and 10% smaller under the BA cartel than under Bertrand-Nash, because banks pass through less of the tax, and the confidence intervals of both differences exclude zero. The loss in bank surplus is instead 15% larger under the Top-4 cartel and 48% larger under the BA cartel, both signif-

[42] See https://www.sanders.senate.gov/wp-content/uploads/inclusive-prosperity-fact-sheet-2019.pdf and https://www.eumonitor.eu/9353000/1/j9vvik7m1c3gyxp/vj75tbvif1yr, respectively, accessed November 1, 2024.

icantly so. Because cartel banks have lower marginal costs, they earn higher per-unit profits before the tax, so each unit of lost quantity entails a larger profit loss. When bank profits are evaluated at Bertrand-Nash marginal costs, the bank-surplus loss under either cartel is not significantly different from Bertrand-Nash, so the additional loss consists of forgone supra-competitive margins, which are significantly different from zero under every conduct model. Finally, tax revenue is 3% higher under the Top-4 cartel and 6% higher under the BA cartel because equilibrium quantities fall by less.

[Place Table 12 here.]

Panel B reports incidence and excess burden per dollar of revenue. Under Bertrand-Nash, incidence is 0.99, so borrowers and banks share the burden roughly equally. Internalization shifts the burden toward banks, and the shift grows with both the degree of internalization and the breadth of the cartel: incidence falls to 0.97 and 0.87 under the Top-4 cartel at $\kappa = 0.2$ and $\kappa = 1$, and to 0.94 and 0.67 under the BA cartel. Under the BA cartel at $\kappa = 1$, banks bear roughly 50% more of the burden than borrowers. The confidence intervals of all four declines exclude zero (Internet Appendix Table H1). The share of revenue borne by borrowers ($-\Delta CS/\Delta\text{Revenue}$) falls as incidence does, from 0.93 to 0.88 under the BA cartel at $\kappa = 1$.

Overall, the tax is highly distortionary. Gross excess burden (inclusive of supra-competitive profits) rises with internalization, from 0.82 under Bertrand-Nash to between 0.83 and 1.10. Because gross excess burden has no predicted sign, the relevant test is two-sided, and the 90% intervals exclude zero for the BA cartel at both degrees of internalization and for the Top-4 cartel at $\kappa = 1$.[43]

Crucially, however, the treatment of supra-competitive profits matters for welfare evaluation. If the policymaker places no social weight on supra-competitive profits, net excess burden moves the other way, falling to 0.79 under the Top-4 cartel and to 0.72 under the BA cartel at $\kappa = 1$ (12% below Bertrand-Nash), though only the difference for the BA cartel at $\kappa = 1$ is statistically significant. The flip in net excess burden results imply that a large share of the distortions come from changes to the supply-driven supra-competitive profits as a response to the tax, rather than from borrower surplus.

Discussion of policy implications. Our results are consistent with the theoretical predictions of Weyl and Fabinger (2013) on the effects of conduct on incidence.[44] From a policy per-

[43]The 90% bias-corrected intervals of the differences in gross excess burden, constructed as in Internet Appendix H, are $[-0.01, 0.05]$ and $[0.01, 0.13]$ for the Top-4 cartel at $\kappa = 0.2$ and $\kappa = 1$, and $[0.02, 0.11]$ and $[0.11, 0.40]$ for the BA cartel.

[44]Ganapati et al. (2020) and O'Connell and Smith (2024) also find that cost-shock incidence on consumers is lower when firms have market power, in the contexts of energy input costs for U.S. manufacturers and U.S. sin

spective, this distinction is important because the policymaker may weigh borrower surplus differently than bank surplus. Our results therefore imply that the desired distributive effects of taxation will depend on the prevailing degree of profit internalization in the market. These findings are also directly relevant to policymakers making hard tax and budget tradeoffs in far from first-best environments where optimal taxation is infeasible.

Furthermore, the treatment of supra-competitive profits is central to whether financial taxation is a desirable policy tool. Benchmarked against estimates from other settings, the net excess burden of the loan tax exceeds that of U.S. retail product taxes (Kroft et al. (2024)) and is close to estimates for income taxes on the top one percent in the U.S. (Saez et al. (2012)) and property transaction taxes in the U.K. (Best and Kleven (2018)), with gross excess burden exceeding all three. A policymaker who naively assumes Bertrand-Nash, or who correctly accounts for conduct but values forgone supra-competitive margins on par with the markups that arise from demand-side market power, would rank the loan tax among the most distortionary instruments available. A policymaker who accounts for conduct and places no social weight on those margins would instead rank it alongside widely used taxes on top incomes and property transactions.

6.1 Heterogeneity in Incidence and Welfare Effects

From a policy perspective, the distributional effects of taxation across borrower types are a first-order concern (Saez and Zucman (2026)). We exploit our micro-level data to examine how the incidence and welfare costs of the financial tax vary with borrower characteristics, illustrating with the alleged BA cartel with $\kappa = 1$.

Figure 4 shows the point estimates by firm size. Under the alleged BA cartel, borrower incidence (Panel A) weakly decreases in firm size (measured by assets), indicating that banks absorb more of the tax burden for larger borrowers, particularly for mid-sized firms. Under Bertrand-Nash, the relationship reverses. A policymaker assuming competitive conduct would predict that banks absorb less of the burden for large borrowers. Beyond the sign difference, the Bertrand-Nash assumption also understates the level of bank absorption across the entire firm-size distribution.

[Place Figure 4 here.]

For (net) excess burden (Panel B), both models agree that distortions decrease with firm

taxes, respectively. Our contribution is to show that this applies not only to demand-based market power but also to supply-side, conduct-based market power.

size. However, as with incidence, assuming Bertrand-Nash competition overstates the level of net excess burden at every point in the firm-size distribution.

While heterogeneity exists, it is modest in magnitude. Under the cartel, moving from the 25th to the 75th percentile of assets is associated with a one-percentage-point decrease in excess burden and a two-percentage-point decrease in incidence.

6.2 Government Subsidies

Having studied taxation, we can readily recover the effects of government subsidies as a mirror-image of the results reported in Table 12. In our simulated model, the tax enters as an additive shifter of marginal cost, so a subsidy is simply a negative tax applied to the same equilibrium pricing equations. Taking the alleged BA cartel as an illustrative benchmark, the estimates imply that a subsidy is expansionary since prices fall and lending volume rises. The implied government multiplier is \$1.10 of combined borrower and producer surplus per dollar of subsidy spending, against \$0.82 under Bertrand-Nash. Banks capture most of these gains, passing through only a fraction of each subsidy dollar to borrowers. Netting out the increase in supra-competitive profits that the subsidy induces through softened competition, the effective multiplier falls to \$0.72 per dollar, compared to \$0.82 under Bertrand-Nash. A policymaker assuming competitive conduct would therefore overestimate the pass-through of subsidies to borrowers and underestimate the share captured by banks as additional markups.

6.3 Revenue Maximizing Tax Rates and the Laffer Curve

To trace out the Laffer curve, we simulate taxes at rates $\tau \in \{0.2, 0.4, 0.5, 0.6, 0.8, \ldots, 4.0\}$ under the alleged BA cartel ($\kappa = 1$), the alleged Top 4 cartel ($\kappa = 1$), and Bertrand-Nash for the pre-tax years. For each year and tax rate, we compute total tax revenue across borrowers and average across years. Figure 5 plots revenue against the tax rate for both conduct models. The current 0.5% rate is far below the revenue-maximizing level.[45] Revenue rises steeply to approximately 2%, then begins to plateau.

[Place Figure 5 here.]

[45] As a validation exercise, we apply a 0.5% tax rate to total private-bank loan volume (including commercial, mortgage, micro and consumer loans), and estimate a total annual tax revenue of \$117 million for 2015-2017. This total is close to the \$96 million in tax revenue from the SOLCA tax reported by the Tax Authority in 2017. The figure shows that under revenue-maximizing taxation of commercial loans alone, the government could raise as much revenue as it currently collects from the tax applied to all loan products at observed tax rates.

Consistent with Miravete et al. (2018), under both cartel models, market power relative to Bertrand-Nash shifts the revenue-maximizing tax rate in the Laffer curve rightward (the solid vertical lines), indicating greater possible tax rates the less competitive the market. The revenue peak also moves upward. The intuition is that cartel banks internalize a larger share of the tax, so total lending volume is less responsive to the tax and the tax base erodes more slowly. The right axis of Figure 5 quantifies the difference: at the 0.5% rate and observed prices, tax revenue under both cartels exceeds Bertrand-Nash revenue by approximately two percentage points, and the gap widens to around six percentage points at revenue-maximizing rates. Under the bootstrap, this revenue gap is statistically distinguishable from zero near the revenue-maximizing rate, where the 90% band excludes zero, but not at the current 0.5% rate (Figure 5).

7 Conclusion

This paper shows that lender conduct is a first-order determinant of who bears the burden of financial taxes and subsidies and of their welfare effects. Using the unanticipated reintroduction of a loan tax in Ecuador and administrative data on the universe of commercial lending, we develop and estimate a structural model of lending with flexible competitive conduct. We reject differentiated Bertrand-Nash competition, the standard assumption in both the banking and public finance literatures, in favor of pricing behavior in which banks internalize rivals' profits within subsets of lenders. Across the conduct models our tests cannot reject, internalization shifts the tax burden toward banks, raises fiscal capacity, and shows a large share of the distortions come from supra-competitive margins, with effects that grow with the degree of internalization.

This finding carries broad implications. A large literature in banking and public finance evaluates taxes, subsidies, and monetary policy pass-through under maintained assumptions about competition that our evidence calls into question. We show that these assumptions are not innocuous. Imposing Bertrand-Nash pricing overstates the share of the tax borne by borrowers, understates fiscal capacity, overstates the effectiveness of credit subsidies at expanding borrowing, and misstates the excess burden of taxation in a direction that depends on the social weight given to forgone supra-competitive profits. Because concentrated and imperfectly competitive banking sectors are the norm rather than the exception worldwide, these biases are likely relevant well beyond our setting.[46] Accordingly, policymakers who design finan-

[46]Internet Appendix Figure A3 shows that three-bank concentration ratios exceed 45% in every major world region and exceed 75% in all but South Asia. Ecuador's ratio of 73.9% is near the LAC median (Figure A2).

cial taxes or subsidies without accounting for how lenders actually compete risk systematically misjudging both who pays and how much it costs.

More broadly, our paper bridges three literatures that have largely developed in parallel. The industrial organization literature on conduct estimation has lacked applications to markets with selection, in part because standard demand-side instruments are unlikely to satisfy exclusion restrictions when borrower characteristics that shift demand also shift marginal costs through default risk. The banking and public finance literatures, meanwhile, have documented that pass-through varies with market concentration but have not disciplined the underlying model of competition with data. We show that quasi-experimental known cost shocks, such as tax changes, resolve the identification problem in selection markets and allow researchers to test, rather than assume, how firms compete. This approach is directly portable to other policy-relevant settings, including the transmission of monetary policy through imperfectly competitive credit and deposit markets.

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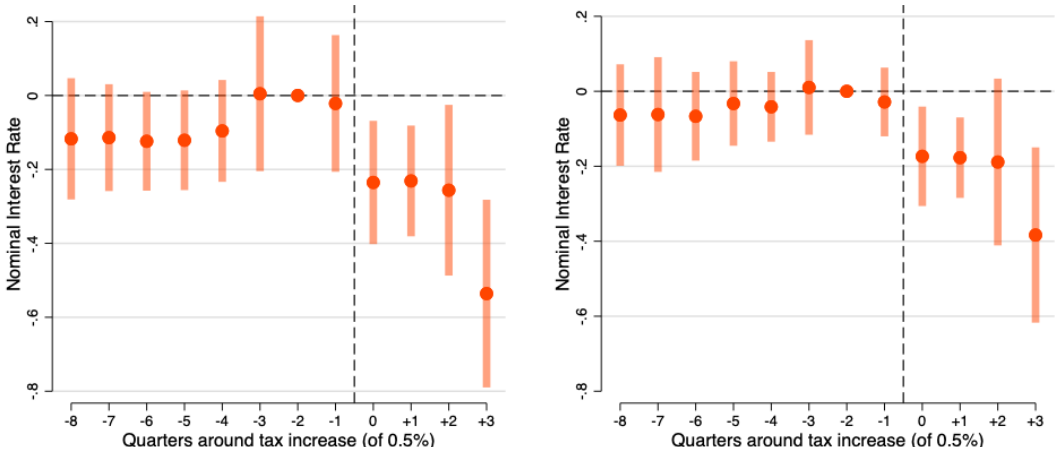
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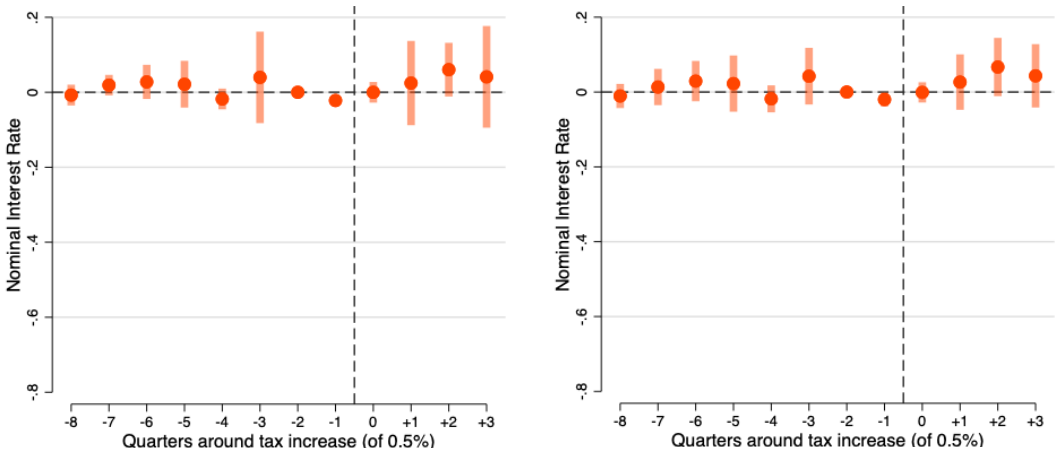
8 Tables and Figures



(a) Regular loans; bank & firm FE (b) Regular loans; bank-firm pair FE

FIGURE 1: EFFECT OF THE TAX ON PRE-TAX NOMINAL INTEREST RATES OF NEW COMMERCIAL CREDIT BY PRIVATE BANKS

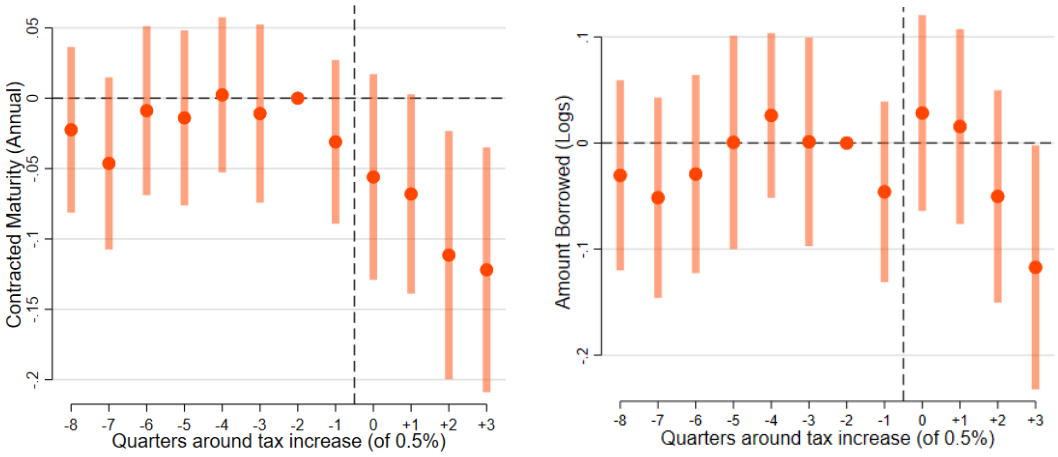
The figure reports the period-by-period difference in average pre-tax nominal interest rates on new commercial loans from private banks around treatment assignment relative to event-time period $t = -2$ (normalized to zero), using firm and bank fixed effects (Panel (a)) or firm $\times$ bank fixed effects (Panel (b)). Data are loan-level. The figure tests for both treatment effects and for evidence of significant differences in outcomes before treatment assignment (pre-trends). Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level.



(a) Regular loans; bank & firm FE (b) Regular loans; bank-firm pair FE

FIGURE 2: EFFECT OF THE TAX ON PRE-TAX NOMINAL INTEREST RATES OF NEW COMMERCIAL CREDIT BY STATE-OWNED BANKS

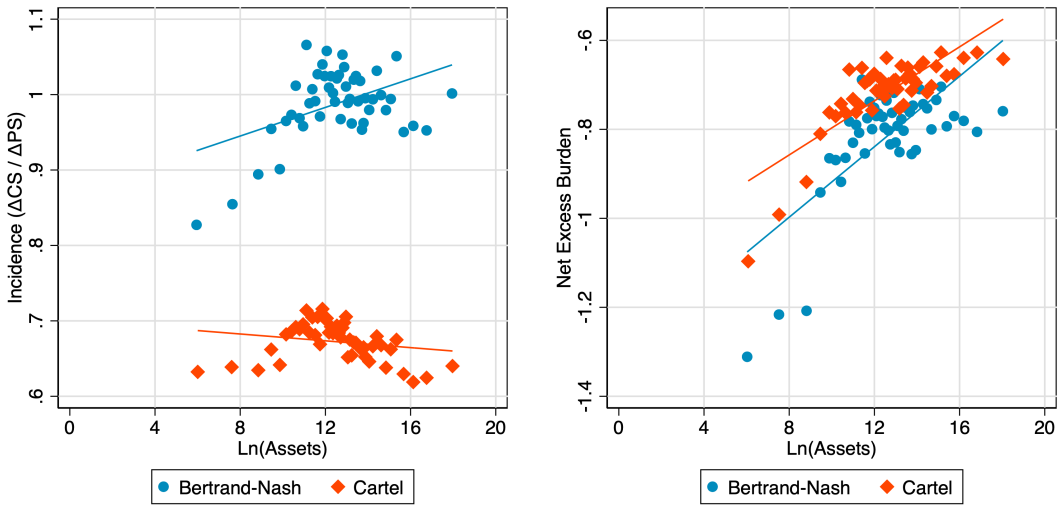
The figure reports the period-by-period difference in average pre-tax nominal interest rates on new commercial loans from public banks around treatment assignment relative to event-time period $t = -2$ (normalized to zero), using firm and bank fixed effects (Panel (a)) or firm $\times$ bank fixed effects (Panel (b)). Data are loan-level. The figure tests for both treatment effects and for evidence of significant differences in outcomes before treatment assignment (pre-trends). Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level.



(a) Regular loans, maturity; bank & firm FE (b) Regular loans, amount; bank & firm FE

FIGURE 3: EFFECT OF THE TAX ON MATURITY AND AMOUNT OF NEW COMMERCIAL CREDIT BY PRIVATE BANKS

The figure reports the period-by-period difference in average term-to-maturity (Panel (a)) or the natural logarithm of the amount borrowed (Panel (b)) on new commercial loans from private banks around treatment assignment relative to event-time period $t = -2$ (normalized to zero), using firm $\times$ bank fixed effects. Data are loan-level. Standard error bars are shown at the 95% confidence level and are clustered at the bank-quarter level.



(a) Incidence

(b) Deadweight Loss

FIGURE 4: HETEROGENEITY IN INCIDENCE AND DEADWEIGHT LOSS BY FIRM SIZE (ASSETS)

The figure reports binscatter plots on the incidence and deadweight loss estimates by firm size (ln assets) by model.

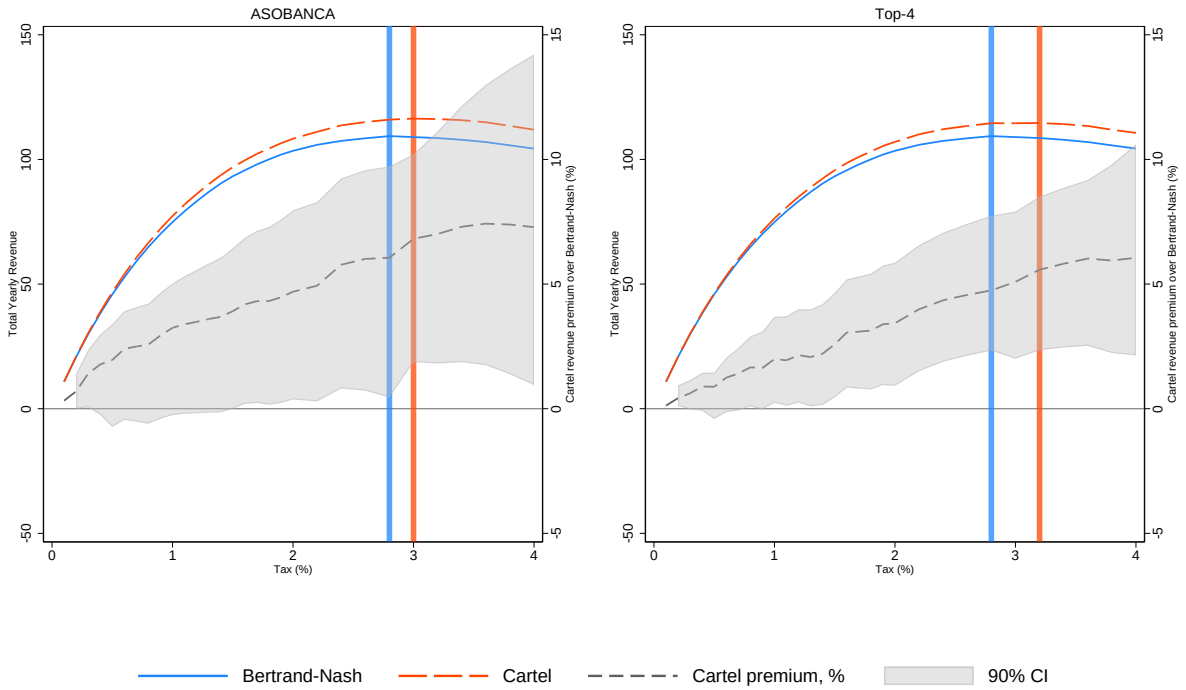


FIGURE 5: TAX REVENUE BY TAX RATE AND MODE OF CONDUCT

The figure reports average tax revenue across all borrowers over tax rates from 0.2 to 4.0%, under Bertrand-Nash and under the panel's alleged cartel (left axis). The left panel defines the alleged cartel as BA members and the right panel as the four largest banks by assets, both at full internalization. The vertical solid lines highlight the revenue-maximizing tax rate under each conduct. The gray dashed line is the cartel revenue premium over Bertrand-Nash (right axis). The shaded band is a 90% confidence region for that premium, formed from the 5th to 95th percentile of the parametric bootstrap distribution over 200 draws of demand parameters and recentered on the plotted premium. The band is computed at tax rates 0.2, 0.5, 1, 2, 2.8, 3, 3.6, and 4%, and interpolated linearly between these eight anchors.

TABLE 1: AGGREGATE-LEVEL CREDIT CHARACTERISTICS

The table describes the commercial loan market in aggregate. Data are at the bank-province-year level for the period 2010 to 2017. Data are for both private and state-owned banks. Bank-province-years with no new loans are excluded. *Total Volume* is the sum of the dollar value of all loans extended in each province-year. *# Clients* is the count of unique corporate clients and *# Loans* is the count of commercial loans extended in each province-year.

Variable	**Mean**	**Median**
Total Volume (2010 USD)	57,323,223	1,400,573
# Clients	83.47	12.00
# Loans	487.38	24.00
Observations	1,976	1,976

TABLE 2: CREDIT CHARACTERISTICS BY MARKET CONCENTRATION (HHI)

The table describes the commercial loan market by market concentration. Data are at the bank-province-year level for 2010 to 2017. Province is defined by the firm's registered location (RUC). State-owned banks and bank-province-years for private banks where no new loans were granted are excluded. HHI is computed using all banks (public and private) based on loan volumes at the province-year level. Data are cut above and below the median HHI value (2,172.98), measured across all bank-province-years in the data. Cell entries report the mean across bank-province-years. All measurements expressed in currency are in 2010 U.S. dollars. Panel A presents branch information. *# Branches* is the number of loan-granting branches in the province. *# Other Private Banks* is the number of other private banks active in the province. *# Other Private Branches* is the total number of loan-granting competing branches active in the province. Panel B presents credit information. *Total Volume* is the sum of the dollar value of all loans extended in each province-year. *# Clients* is the count of unique clients and *# Loans* is the count of loans extended in each province-year. *Av. Loan* is the average loan size. *Av. Maturity* is the average annualized term-to-maturity at issuance. *Av. Interest Rate* is the volume-weighted nominal, annualized interest rate at issuance, in percent. *# Loans per Client* is the average number of loans extended per firm from a given bank. *Default Rate* is the share of loans that default over their term, in percent, excluding loans with maturity extending past 2017.

Variable	**Below Median HHI**	**Above Median HHI**
	Panel A: Branch Information	
# Branches	5.12	2.90
# Other Private Banks	16.43	10.48
# Other Private Branches	103.33	46.32
Observations	818	836
	Panel B: Credit Information	
Total Volume (M 2010 USD)	82.73	29.44
# Clients	123.00	36.17
# Loans	903.07	164.21
Av. Loan (K 2010 USD)	177.90	98.06
Av. Maturity	1.13	0.88
Av. Interest Rate (%)	9.82	10.75
# Loans per Client	5.54	3.66
Default Rate (%)	0.40	0.42
Observations	818	836

TABLE 3: DESCRIPTIVE STATISTICS

The table describes the commercial loan dataset for 2010–2017. Panel A summarizes data for firm-years with borrowing and Panel B for firm-years without borrowing. *Firm Age* is years since incorporation. *Total Wages* are all wages reported to the company regulator for both contract and full-time employees. *Total Debt* is the sum of short- and long-term debt. *Leverage* is total debt over beginning-of-period total assets. *Number Bank Relationships* is the number of banks the firm borrows from in a year. *Number Loans* is the number of loans the firm borrows in a year. Panel C summarizes firm-bank-year data. *Age Bank Relationship* is years since the first loan with a bank. Panel D summarizes loan-year level data for commercial loans at origination. *Interest Rate* is the nominal, annualized interest rate at issuance, in percent. *Loan Amount* is the size of the loan at issuance. *Annual Loan Maturity* is years-to-maturity at issuance. *1(Loan with rating < B)* equals one if the bank assigns a risk weight below B, i.e., non-zero expected write-down at issuance. *1(Default Observed)* equals one if the loan defaults at any point after issuance. Panel E summarizes bank-year data. *Deposit Interest Rates* is a weighted average of bank-year deposits, where weights are the nationwide average rates for deposits at each horizon. All monetary values are in millions of 2010 U.S. dollars. Continuous variables are winsorized at 1% and 99%.

Variable	Mean	Median	SD	Min.	Max.	Obs.
	Panel A: Firm-Level Data: Active Borrowers					
Firm Age	12.07	9.00	11.00	0.00	96.00	92,659
Total Assets (M 2010 USD)	1.93	0.39	4.06	0.00	20.66	92,659
Total Sales (M 2010 USD)	2.43	0.60	4.68	0.00	23.14	92,659
Total Wages (M 2010 USD)	0.34	0.10	0.61	0.00	2.98	92,659
Total Debt (M 2010 USD)	1.24	0.27	2.53	0.00	12.65	92,659
Leverage	0.67	0.71	0.28	0.00	1.19	92,659
Number Bank Relationships	1.40	1.00	0.80	1.00	7.00	92,659
Number Loans	9.23	2.00	103.39	1.00	9,195.00	92,659
	Panel B: Firm-Level Data: Non Active Borrowers					
Firm Age	9.92	7.00	10.09	0.00	93.00	359,827
Total Assets (M 2010 USD)	0.46	0.05	1.73	0.00	20.66	359,827
Total Sales (M 2010 USD)	0.43	0.03	1.70	0.00	23.14	359,827
Total Wages (M 2010 USD)	0.07	0.01	0.25	0.00	2.98	359,827
Total Debt (M 2010 USD)	0.26	0.02	1.01	0.00	12.65	359,827
Leverage	0.54	0.58	0.40	0.00	1.19	359,827
	Panel C: Relationship-Level Data					
Age Bank Relationship	2.29	2.00	2.40	0.00	16.00	127,909
	Panel D: Loan-Level Data					
Loan Interest Rate	9.83	9.02	2.59	0.01	25.50	828,279
Loan Amount (M 2010 USD)	0.07	0.01	0.22	0.00	2.00	828,279
Annual Loan Maturity	0.50	0.25	0.81	0.00	17.84	828,279
1(Loan with Rating < B)	0.02	0.00	0.14	0.00	1.00	828,279
1(Default Observed)	0.00	0.00	0.06	0.00	1.00	694,514
	Panel E: Bank-Level Data					
Deposit Interest Rates	4.73	4.59	0.49	3.89	6.26	173

TABLE 4: AGGREGATE PASS-THROUGH ESTIMATES

The table reports aggregate pass-through estimates to the tax-inclusive interest rates of commercial loans around the introduction of the 2014 SOLCA tax in Ecuador. Data are at the loan-level. The estimation window is from eight quarters before the introduction of the tax through three quarters afterward, excluding October 2014. The main independent variable is the tax rate, measured as 0.5 adjusted proportionally by term-to-maturities if maturity is less than 1 year. The dependent variable is the tax-inclusive interest rate, which is the sum of the nominal, annualized interest rate plus the tax rate. Both are in percentage points. Regressions control for twenty buckets of term-to-maturity, and twenty buckets of loan amount. Models (2) and (4) control for predicted default probability. Models (1) and (2) control for bank and firm fixed effects, whereas (3) and (4) control for bank × firm (pair) fixed effects. Robust standard errors clustered at the bank-quarter level are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Testing is conducted against the full pass-through null hypothesis (PT = 1).

	Outcome: Tax-inclusive interest rate			
	(1)	(2)	(3)	(4)
Pass-through ($\widehat{\text{PT}}$)	0.358***	0.334***	0.531***	0.540***
	(0.144)	(0.166)	(0.137)	(0.150)
Pr(Default) Control	No	Yes	No	Yes
Maturity & Amount Controls	Yes	Yes	Yes	Yes
Bank Fixed Effect	Yes	Yes	No	No
Firm Fixed Effect	Yes	Yes	No	No
Pair Fixed Effect	No	No	Yes	Yes
Observations	385,128	352,574	378,747	347,471
R-squared	0.721	0.711	0.783	0.777

TABLE 5: HETEROGENEITY IN DIRECT TAX PASS-THROUGH

This table reports heterogeneity in aggregate pass-through estimates to the tax-inclusive interest rates of commercial loans around the introduction of the 2014 SOLCA tax in Ecuador. Data are at the loan level for 2010–2017, excluding October 2014. The main independent variable is the tax rate, measured as 0.5 adjusted proportionally by term to maturity if maturity is less than one year. The dependent variable is the tax-inclusive interest rate (nominal annualized rate plus tax rate), in percentage points. Regressions control for twenty buckets each of term to maturity, loan amount, and predicted default probability, as well as bank × firm (pair) fixed effects. Interacted variables are: *# Potential Lenders*, the maximum number of active lenders as of October 2014; *City HHI*, the Herfindahl–Hirschman Index per city-year prior to October 2014; *Multi-market Contact*, the average number of other provinces in which bank pairs jointly operate; and *Market Share ASOBANCA Members*, the loan-weighted market share of ASOBANCA (BA) members in the market. All interacted variables are standardized so the main effect is the pass-through for the average borrower. Robust standard errors clustered at the bank-quarter level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. For the main effect, testing is against the full-pass-through null (PT = 1). For interactions, testing is against zero.

Outcome: Tax-inclusive interest rate				
	(1)	(2)	(3)	(4)
Pass-through ($\widehat{\text{PT}}$)	0.446***	0.606**	0.532**	0.529**
	(0.207)	(0.200)	(0.203)	(0.203)
Interaction term:				
	# Potential Lenders	*City HHI*	*Multimarket Contact*	*Mkt. Share ASOBANCA*
	0.458***	−0.410***	−0.225**	−0.241**
	(0.117)	(0.087)	(0.092)	(0.116)
Pair fixed effect	Yes	Yes	Yes	Yes
Amount bucket	Yes	Yes	Yes	Yes
Maturity bucket	Yes	Yes	Yes	Yes
Default risk control	Yes	Yes	Yes	Yes
Observations	347,471	347,463	347,471	347,471
R^2	0.777	0.777	0.777	0.777

TABLE 6: DEMAND PARAMETERS: DISCRETE AND CONTINUOUS ESTIMATES

This table presents estimated parameters for the discrete and continuous choice models. Standard errors are in parentheses. Interest rate coefficients are estimated via the instrumental variable approach in Equation 26 (first-stage F-statistic: 11.08; overidentification p-values: 0.77 discrete, 0.73 continuous). All other coefficients and standard errors are means across regions. Region-level estimates are reported in Internet Appendix D.

	Discrete	**Continuous**
Interest Rate (α)	−0.599 (0.310)	−0.488 (0.129)
Age	−0.020 (0.303)	0.076 (0.105)
Assets	0.554 (1.336)	7.607 (2.292)
Debt	−0.178 (1.929)	−1.380 (1.923)
Capital	0.410 (0.522)	−0.255 (0.570)
Revenue	−1.961 (1.914)	−0.218 (0.685)
Wages	2.588 (1.188)	−0.617 (0.711)
Expenditures	1.357 (1.323)	0.205 (0.741)
Age Relationship	0.336 (0.048)	0.186 (0.047)
Interactions with Interest Rate		
Age × Rate	−0.039 (0.022)	−0.006 (0.012)
Assets × Rate	−0.103 (0.206)	0.176 (0.162)
Debt × Rate	0.019 (0.041)	−0.032 (0.132)
Capital × Rate	−0.008 (0.091)	−0.051 (0.047)
Revenue × Rate	0.075 (0.038)	0.022 (0.037)
Wages × Rate	0.154 (0.125)	0.034 (0.060)
Expenditures × Rate	−0.038 (0.098)	0.012 (0.044)
Age Rel. × Rate	0.001 (0.003)	−0.011 (0.004)
σ_α (Interest Rate)	0.402 (0.144)	
σ_ψ (Constant)	0.988 (0.010)	
σ_ε^l (Continuous)		1.807 (0.034)

TABLE 7: LOAN DEMAND: OWN-PRODUCT AND CROSS-PRODUCT ELASTICITIES

This table reports loan-level estimated elasticities for realized and non-realized loans. Continuous elasticity is the intensive-margin elasticity with respect to interest rates. Discrete elasticity is the extensive-margin elasticity with respect to interest rates. Total is the sum of continuous and discrete. Cross elasticity is the discrete bank-substitution elasticity with respect to interest rates, averaged at the pair level over all other alternatives in the choice set.

Elasticities	Mean	Median	Std. Dev.	Min.	Max.	Count
Continuous	−3.45	−3.39	1.28	−6.98	0.00	307,168
Discrete	−4.76	−5.01	2.61	−10.84	0.00	307,168
Total	−8.21	−8.25	3.36	−16.53	0.00	307,168
Cross	0.62	0.12	1.22	−0.14	6.40	307,168

TABLE 8: DESCRIPTION OF MODEL FIT

The table presents measures of model fit regarding market shares, loan use, prices, and default rates. Differences in observations are because loan use, prices, and default are only measured for actual, realized loans. Market shares and loan use come from the structural demand model, discussed in Section 4. Estimation methodologies for default and prices are available in Internet Appendix C.

Parameter	Mean	Std. Dev.	Count
Observed Market Share	0.12	0.32	368,257
Model Market Share	0.12	0.21	368,257
Observed Loan Use	9.25	2.43	39,324
Predicted Loan Use	9.14	2.30	39,356
Observed Prices	11.44	4.38	39,356
Predicted Prices	11.09	3.96	39,356
Observed Default	0.02	0.15	39,356
Predicted Default	0.02	0.03	39,356

TABLE 9: TESTING BERTRAND-NASH AGAINST ALTERNATIVE CARTEL MODELS

This table tests Bertrand-Nash ($\kappa = 0$) against candidate cartel models using the Rivers-Vuong model-selection test with the effective F diagnostic of Duarte et al. (2024). Candidates vary along two dimensions: cartel membership and internalization degree (κ). Panel A defines the alleged cartel as ASOBANCA members, Panel B as the four largest banks by assets, and Panel C as all banks in the market. A positive T^{RV} favors the alternative over Bertrand-Nash. The excluded instrument is the loan-level effective tax (maturity × rate × private-lender indicator), its square, and the public and cartel interactions. Column 4 reports the effective F, Column 5 the critical value it must exceed for best-case power to reach 95 percent, and Column 6 the degeneracy measure ρ^2, which together with the number of instruments indexes the critical value. Worst-case size is bounded at 7.5 percent at every effective F in every comparison, so the corresponding size critical value is zero throughout and is not reported.

Panel A. Bertrand-Nash vs. Banking Association Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.0	0.2	4.90	15.48	5.7	0.142
0.0	0.4	4.77	15.85	5.8	0.131
0.0	0.6	4.46	16.30	5.9	0.114
0.0	0.8	4.44	15.74	6.5	0.070
0.0	1.0	5.21	14.48	7.1	0.029

Panel B. Bertrand-Nash vs. Top 4 Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.0	0.2	5.02	15.18	5.7	0.141
0.0	0.4	5.02	15.13	5.7	0.140
0.0	0.6	5.03	15.09	5.8	0.133
0.0	0.8	5.06	14.88	5.8	0.128
0.0	1.0	5.18	14.92	5.8	0.130

Panel C. Bertrand-Nash vs. Industry-Wide Cartel

Null (κ) (1)	Alternative (κ) (2)	T^{RV} (3)	Effective F (4)	Critical Value (5)	ρ^2 (6)
0.0	0.2	4.86	20.75	7.1	0.112
0.0	0.4	4.28	22.82	7.7	0.073
0.0	0.6	3.03	24.72	8.8	0.012
0.0	0.8	3.48	23.16	9.4	0.001
0.0	1.0	4.10	21.39	9.0	0.009

TABLE 10: MODEL CONFIDENCE SET

This table reports model confidence set p-values for all candidate conduct models. Rows index the conduct model and columns the degree of profit internalization κ; Bertrand–Nash corresponds to $\kappa = 0$. The procedure is a single sequential elimination: it tests whether the models still under consideration fit equally well, drops the worst-fitting model when equal fit is rejected, and repeats until it can no longer reject. The reported p-value is the significance level at which a model is eliminated, so ranking entries from smallest to largest recovers the elimination order.

	Degree of internalization κ					
Conduct model	0.0	0.2	0.4	0.6	0.8	1.0
Bertrand–Nash	0.000					
Top-4 banks		0.271	0.265	0.260	0.256	1.000
Banking Association		0.138	0.105	0.141	0.298	0.716
Industry-wide		0.066	0.041	0.050	0.019	0.002

TABLE 11: SUPPLY-SIDE ESTIMATES: MARGINAL COSTS AND MARKUPS

This table reports borrower-group-bank specific supply-side parameters under alternative competitive conduct. Each panel corresponds to a conduct model. Rows report marginal costs, markups, and Lerner indices (mean/median).

	Mean	**Median**
Panel A: Bertrand–Nash Competition		
Marginal Cost	5.95	6.77
Markup	2.98	3.21
Lerner Index	0.50	0.44
Panel B: BA Cartel		
Marginal Cost	4.96	5.94
Markup	3.97	4.09
Lerner Index	0.68	0.52
Panel C: Top-4 Bank Cartel		
Marginal Cost	5.76	6.60
Markup	3.16	3.44
Lerner Index	0.52	0.45
Panel D: Industry-Wide Cartel		
Marginal Cost	5.59	5.82
Markup	3.33	4.12
Lerner Index	0.75	0.55

TABLE 12: TAX INCIDENCE AND EXCESS BURDEN ACROSS CONDUCT SPECIFICATIONS

This table reports simulated tax incidence and excess burden from introducing a uniform loan tax of $\tau = 0.5\%$ under five conduct regimes: Bertrand-Nash, Top-4 internalization at $\kappa = 0.2$ and $\kappa = 1$, and BA internalization at $\kappa = 0.2$ and $\kappa = 1$. Panel A reports medians of firm-year changes in borrower surplus, bank surplus, and tax revenue. Net bank surplus evaluates profits using the Bertrand-Nash marginal-cost schedule as a common baseline; the marginal-cost row reports the associated adjustment. Panel B reports means of incidence $\Delta CS/\Delta PS$, gross excess burden $-(\Delta CS + \Delta PS + \Delta\text{Revenue})/\Delta\text{Revenue}$, net excess burden using Bertrand-Nash-cost-adjusted bank surplus, and borrower burden per dollar of revenue $-\Delta CS/\Delta\text{Revenue}$. Positive excess burden denotes a welfare loss. The sample stacks five regions, restricts attention to years before 2015 and realized lending relationships, and trims each outcome at its 10th and 90th percentiles within conduct specification before calculating medians and means. Parentheses report parametric bootstrap standard errors, calculated as the standard deviation across 500 draws of demand parameters. Brackets report 80% bias-corrected bootstrap intervals.

	Bertrand-Nash	Top4 κ=0.2	Top4 κ=1	BA κ=0.2	BA κ=1
Panel A: Medians of Δ Outcomes					
Δ Borrower surplus (CS)	-4,432.46	-4,427.05	-4,250.76	-4,441.81	-3,990.42
	(1,284.64)	(1,253.95)	(1,241.18)	(1,254.95)	(1,273.89)
	[-4,841, -1,462]	[-4,855, -1,707]	[-4,787, -1,569]	[-4,879, -1,670]	[-4,585, -1,082]
Δ Bank surplus (PS)	-4,657.98	-4,782.42	-5,363.75	-4,971.18	-6,911.78
	(1,282.04)	(1,288.19)	(1,262.43)	(1,267.54)	(990.10)
	[-5,747, -2,619]	[-5,819, -2,756]	[-6,186, -3,385]	[-5,974, -2,910]	[-7,217, -5,352]
Δ Tax revenue	5,334.47	5,346.03	5,485.18	5,349.91	5,665.25
	(436.64)	(438.39)	(436.81)	(436.94)	(437.37)
	[4,819, 5,978]	[4,832, 5,988]	[4,967, 6,118]	[4,834, 5,984]	[5,148, 6,290]
Δ PS (efficiency component)	0.00	-2.42	-16.38	-42.02	-317.43
	(-)	(0.71)	(5.18)	(13.85)	(127.09)
	[0, 0]	[-3, -1]	[-24, -11]	[-52, -16]	[-480, -144]
Δ Bank surplus (net)	-4,657.98	-4,670.21	-4,658.45	-4,668.95	-4,574.05
	(1,282.04)	(1,281.44)	(1,219.68)	(1,242.90)	(1,022.43)
	[-5,747, -2,619]	[-5,725, -2,660]	[-5,718, -2,663]	[-5,716, -2,649]	[-5,704, -3,097]
Panel B: Means of Incidence and DWL					
Incidence (ΔCS / ΔPS)	0.99	0.97	0.87	0.94	0.67
	(0.34)	(0.31)	(0.21)	(0.28)	(0.22)
	[0.71, 1.40]	[0.72, 1.33]	[0.66, 1.07]	[0.70, 1.26]	[0.39, 0.74]
Excess Burden	0.82	0.83	0.90	0.87	1.10
	(0.40)	(0.41)	(0.41)	(0.39)	(0.39)
	[0.29, 1.11]	[0.31, 1.12]	[0.36, 1.17]	[0.34, 1.15]	[0.56, 1.35]
Net Excess Burden	0.82	0.82	0.79	0.83	0.72
	(0.40)	(0.41)	(0.39)	(0.39)	(0.37)
	[0.29, 1.11]	[0.29, 1.11]	[0.25, 1.08]	[0.32, 1.12]	[0.23, 1.04]
Δ CS / Δ Revenue	0.93	0.93	0.91	0.93	0.88
	(0.26)	(0.26)	(0.26)	(0.26)	(0.27)
	[0.55, 1.07]	[0.55, 1.06]	[0.53, 1.05]	[0.55, 1.06]	[0.50, 1.04]